

Factual Recall Circuit Analysis in Gemma-2-2B

Abstract

We report a mechanistic interpretability analysis of factual recall reasoning in Gemma-2-2B, using Neuronpedia attribution graphs. We generated 10 attribution graphs across prompts spanning geography (Nigeria, Ghana, Nile, Everest), history (Napoleon, US president), science (water composition, cell biology, relativity), and literature (Hamlet), and identified 222 recurring circuit components at a 50% co-occurrence threshold (features present in $\geq 5/10$ graphs). The top recurring feature (feature 110446948, layer 4, node_id=4_14857_2, appearances=34/10) is labelled "code snippets and license agreements", indicating that the shared factual-recall circuit is dominated by general-register and syntactic features rather than domain-specific knowledge encoders. The most domain-specific features appear in late layers (L16–L25): "cell biology" (feature=1578, L16), "rivers and watersheds" (feature at L19), "geographic locations, especially islands" (L18, act=51.48 for Napoleon), and "US president/Obama/Trump" (feature=318780, L20). Prediction confidence varied dramatically across prompts: water/oxygen achieved 97.8% confidence (most over-trained fact), while Nigeria, US president, and Mount Everest fell below 30%. Dominant causal paths were predominantly direct (embedding→logit) or short multi-hop chains through late layers. The inhibitory path in the Napoleon/Elba circuit (L6→L25→logit) provides the clearest example of token competition suppression. We find that factual recall circuit confidence correlates with the frequency and specificity of the fact in training data, and that the recurring structural circuit is largely domain-agnostic, with domain knowledge encoded in long-tail late-layer features that activate only when the specific fact is highly represented in training.

Introduction

This paper presents a mechanistic interpretability analysis of how Gemma-2-2B implements factual recall reasoning. Using attribution graph analysis via the Neuronpedia platform, we generated and analysed 10 attribution graphs, one per factual recall prompt, and identified the shared circuit components that recur across this category of reasoning.

Factual recall prompts require the model to complete statements whose answers are well-defined facts stored in world knowledge (e.g., "The capital of Nigeria is", "Hamlet was written by"). Unlike analogical or linguistic reasoning, factual recall places maximum demand on entity and fact retrieval: the model must identify a specific named entity or categorical fact from a broad search space with minimal structural cues in the prompt itself.

Our analysis proceeds in three stages: (1) cross-graph pattern mining to identify recurring circuit components shared across all 10 prompts; (2) edge-level neighbourhood analysis of the top recurring features to understand their wiring; and (3) per-prompt causal path tracing to reconstruct the exact computational pathway from input to output for each prompt.

Methods

Model: Gemma-2-2B

SAE: gemmascope-transcoder-16k

Prompts: 10 factual recall prompts spanning geography, history, science, and literature

Graph generation: Neuronpedia attribution graph API with maxFeatureNodes=3000, desiredLogitProb=0.95, nodeThreshold=0.8, edgeThreshold=0.85

Cross-graph threshold: 50% (feature must appear in $\geq 5/10$ graphs to qualify as recurring)

Per-prompt analysis: top-40 nodes labeled via Neuronpedia SAE feature API, causal paths traced via greedy edge-weight walk to logit node (max 5 paths, max 6 hops, min edge weight 0.05)

Results

Recurring Features (50% threshold — appears in ≥5/10 graphs)

Total recurring features found: 222

Top-15 recurring features by (appearances, avg_influence):

Layer	Feature	Node_id	Appearances	Avg Influence	Label
4	110446948	4_14857_2	34/10	0.6152	code snippets and license agreements
6	2586668	6_2267_4	32/10	0.648	words that appear in programming code, legal jargon, or scientific texts
3	5150441	3_3205_2	24/10	0.639	code snippets and documentation references, possibly related to web development
7	4828270	7_3099_2	22/10	0.7077	a variety of reference codes, abbreviations, and identifiers from different fields
0	40747877	0_9026_2	21/10	0.5454	technical documents or data, including numbers, units, and references to figures or tables
4	47653198	4_9757_1	20/10	0.6173	various acronyms, IDs, and symbols, possibly related to scientific data
0	2239785	0_2115_2	20/10	0.5995	data reported as a percentage inside brackets, especially in a laboratory or medical context, and also recognizes countries
5	2584395	5_2267_5	19/10	0.6766	law related terminology and references to specific cases or legal entities
0	11637899	0_4823_1	19/10	0.6524	the word "part" followed by prepositions or words related to sections or components
0	74438300	0_12200_1	19/10	0.6433	a variety of specific nouns

0	1708475	0_1847_2	19/10	0.5713	scientific terms and experimental details related to biological and chemical research
0	70051365	0_11835_2	18/10	0.6317	terms used in software code such as "assembly", "using", "namespace", and "license"
1	99962728	1_14137_1	16/10	0.5813	words or phrases that appear in legal or technical documents
0	2399144	0_2189_2	15/10	0.7105	technical writing related to scientific studies
0	33329529	0_8163_2	14/10	0.6902	words and phrases related to societal and political issues

Edge Neighbourhood Analysis (Step 5a)

Top-1 Recurring Feature: node_id=4_14857_2 (L4, feature=110446948, appearances=34/10, avg_inf=0.6152)

Label: "code snippets and license agreements"

Incoming edges (top 10) — what feeds INTO this feature:

Source node_id	Layer	Feature	Edge weight
E_2_0	E (embed)	0	-20.1478
E_651_1	E (embed)	1	+10.895
3_3205_2	3	5150441	-9.6127
2_2051_2	2	2110482	-3.8559
E_6037_2	E (embed)	2	-3.7549
0_3_2	0	-1	+3.4006
0_1961_2	0	1925702	+2.8667
2_7173_2	2	25751073	+2.1349
3_13984_2	3	97839062	+2.0001
0_1903_1	0	1813559	+1.9273

Outgoing edges (top 10) — what this feature feeds INTO:

Target node_id	Layer	Feature	Edge weight
7_3099_2	7	4828270	-12.2255

6_9865_2	6	48733121	-9.7844
15_4494_2	15	10172289	-1.0686
6_4662_2	6	10902108	-1.015
16_7171_2	16	25837249	+0.8759
7_691_2	7	244642	-0.6167
5_3333_2	5	5576124	+0.5835
19_9215_5	19	42647210	-0.272
18_15734_5	18	124086362	-0.2175
24_13277_5	24	88478228	-0.1741

Key observation: Feature 4_14857_2 receives the strongest inputs from embedding-layer nodes (E_2_0, E_651_1) and from the Layer 3 feature 3_3205_2 (also a top-3 recurring feature). It sends its largest outputs to Layer 7 (feature 4828270, node 7_3099_2 — the #4 recurring feature) and Layer 6 (feature 48733121), both with strong negative (inhibitory) weights. This suggests it acts as a cross-layer inhibitory gating node coordinating between embedding space and mid-layer features.

Top-2 Recurring Feature: node_id=6_2267_4 (L6, feature=2586668, appearances=32/10, avg_inf=0.648)

Label: "words that appear in programming code, legal jargon, or scientific texts"

Incoming edges (top 10):

Source node_id	Layer	Feature	Edge weight
E_2_0	E (embed)	0	-5.9484
4_14857_4	4	110446948	-2.6729
E_576_3	E (embed)	3	-1.8265
E_15922_4	E (embed)	4	+1.7562
E_6037_2	E (embed)	2	-1.6645
E_651_1	E (embed)	1	+1.2052
3_3205_4	3	5150441	-1.1261
0_15636_4	0	122265702	+1.1108
0_5_4	0	-1	+1.0502
0_3_4	0	-1	+0.6806

Outgoing edges (top 10):

Target node_id	Layer	Feature	Edge weight
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8_3099_4	8	4831377	-11.34
11_3544_4	11	6324334	-11.1059
7_3099_4	7	4828270	-11.0382
14_4649_4	14	10878765	-3.2724
9_11399_4	9	65088335	+1.5289
19_2188_4	19	2438716	+1.3991
8_14523_4	8	105596769	+1.0717
8_9091_4	8	41409541	+0.9368
8_7435_4	8	27710281	+0.9039
16_7449_4	16	27874294	+0.8525

Key observation: Feature 6_2267_4 is a high-fanout node (in_degree=52, out_degree=50). Its strongest outputs are all inhibitory (−11.34, −11.10, −11.04) directed at L8, L11, and L7 features. The L7 target (feature 4828270) is also the #4 recurring feature. This feature acts as a strong inhibitory bottleneck: it receives mixed signals from embeddings and L3-L4, then sends massive inhibitory signals into the middle-layer processing chain.

Hub Layer Analysis (Step 5b)

Hub layer: Layer 0 (appears 7 times in top-15 recurring features; 211 nodes in representative graph)

Layer 0 dominates the recurring features table in terms of raw feature count (7 of the top 15 entries). In the representative graph (Nigeria), Layer 0 contains 211 nodes — the largest layer in the graph.

Top-20 Layer 0 nodes by influence (Nigeria graph):

node_id	Feature	Influence	Activation
0_15038_4	113093279	0.7995	3.0159
0_5319_2	14153859	0.7988	3.3592
0_12215_5	74621435	0.7982	2.3931
0_13763_2	94730729	0.7974	1.7245
0_9773_2	47770424	0.7963	1.9935
0_2146_1	2305877	0.7961	2.365
0_13943_2	97224539	0.7952	1.8181
0_12459_1	77632029	0.795	2.3991
0_8163_2	33329529	0.7944	2.0843
0_12955_2	83935445	0.7929	1.6285

0_15415_2	118834235	0.7926	1.4628
0_9773_1	47770424	0.7913	2.9707
0_8047_5	32389175	0.7903	2.3207
0_11629_2	67634264	0.7892	2.8854
0_3718_3	6917339	0.7885	2.3827
0_52_3	1430	0.7874	2.0893
0_9147_1	41847525	0.7865	2.4207
0_15571_2	121251377	0.7847	2.4311
0_8163_4	33329529	0.7819	2.7023
0_4711_2	11103827	0.7801	1.7846

Interpretation: Layer 0 nodes show uniformly high influence scores (0.78–0.80), consistent activation levels (1.4–3.4), and serve as the broadest input recognition layer — detecting token-level surface features across all token positions (ctx_idx varies from 1 to 5). The diversity of features at L0 reflects that factual recall prompts activate a wide range of surface-form detectors before the circuit narrows to domain-specific processing in middle layers.

Per-prompt Circuit Interpretations

Prompt: "The capital of Nigeria is"

Predicted token: Output " a" (p=0.173) (prob=0.1731)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
4	9467	4_9467_5	0.8001	2.0753	words associated with government, law, and crime
0	15038	0_15038_4	0.7995	3.0159	locations and organizations
1	7200	1_7200_2	0.7992	1.8948	mentions of geographic locations in the UK.
0	5319	0_5319_2	0.7988	3.3592	many different words associated with completely different topics, spanning games, science, sports, visual media, food and community

0	12215	0_12215_5	0.7982	2.3931	the word "is"
4	4104	4_4104_1	0.798	6.9604	words related to study designs, results, and published documents
3	3488	3_3488_3	0.7976	3.3693	words or phrases about locations or things being part of other things, especially about clinics or medical analysis
0	13763	0_13763_2	0.7974	1.7245	the phrase 'country of origin'
7	3099	7_3099_5	0.7969	5.8022	a variety of reference codes, abbreviations, and identifiers from different fields.
1	11207	1_11207_1	0.7965	3.315	code and documented code
0	9773	0_9773_2	0.7963	1.9935	references to dates and times
0	2146	0_2146_1	0.7961	2.365	the word survivor (or a variant) and sometimes a preceding article of 'the'.
6	9577	6_9577_4	0.7957	3.2699	words and phrases related to nationality and patriotism
4	5390	4_5390_5	0.7954	2.3771	phrases containing auxiliary verbs "is," "are," "was". "be" and action verbs often including "to" and "by".
0	13943	0_13943_2	0.7952	1.8181	the word "recent" and words appearing near to it
0	12459	0_12459_1	0.795	2.3991	the word "damage" and sometimes other words near "damage" or related to negative experiences
6	3182	6_3182_5	0.7948	2.7502	mentions of cities or regions
6	16070	6_16070_5	0.7946	3.6775	phrases related to formulas or derivations.
0	8163	0_8163_2	0.7944	2.0843	words and phrases related to societal and political issues
1	1251	1_1251_1	0.7942	3.0339	instances of the word "The"
6	5695	6_5695_4	0.7937	2.8771	mentions of specific corporate entities and product or character names from particular fictional universes, in addition to terms related to marine biology

3	11806	3_11806_5	0.7935	2.3508	words/phrases related to story telling
2	14967	2_14967_5	0.7931	2.3753	words and phrases related to governmental leadership and elections.
0	12955	0_12955_2	0.7929	1.6285	vague positive terms and terms related to rules and laws
0	15415	0_15415_2	0.7926	1.4628	mentions of opinions, facts or reasons
3	10202	3_10202_3	0.7924	3.8018	terms related to code and US immigration
6	11331	6_11331_4	0.7922	4.2509	names of places, mostly countries, provinces and cities
4	9089	4_9089_5	0.792	3.1997	words related to politics, finance and government.
1	6426	1_6426_2	0.7918	2.2059	mentions of genetic sequences, primers, and genes

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
8	9091	8_9091_4	0.7999	5.4144	mentions of a person named "Ilube", possibly along with some titles
9	5986	9_5986_5	0.7997	5.5815	words that indicate a superlative or high importance
8	589	8_589_4	0.799	5.0684	words related to culture, particularly Indian culture, as well as family structure and cuisine.
9	4854	9_4854_5	0.7986	4.5138	mentions of a particular city, and also mentions of "Free Water"
0	10	0_10_2	0.7984	0.0	the keyword 'let' which is used for variable declarations in Javascript
8	5480	8_5480_5	0.7978	3.5821	proper nouns referring to countries or political entities, and words associated with international trade and economics
11	3544	11_3544_4	0.7967	22.4298	scientific or technical words and jargon

10	7276	10_7276_5	0.7959	4.3549	words associated with institutional, professional, and/or academic language
11	12600	11_12600_4	0.7939	10.1481	references to countries or regions
15	4494	15_4494_3	0.7933	31.3888	the word "capital" and sometimes letters

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
24	5668	24_5668_5	0.7971	11.9528	text written in all caps, especially political text

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=13.173149 / hops=1 → node_id=E_603_5 | feature=? | Layer E | (no label) — edge [+13.1731] → **LOGIT** node_id=27_476_5 Layer 27 — Output " a" (p=0.173)

Path 2 — (±) mixed / weight=0.589485 / hops=2 → node_id=9_4854_5 | feature=4854 | Layer 9 | mentions of a particular city, and also mentions of "Free Water" — edge [+0.4579] → node_id=25_9975_5 | feature=? | Layer 25 | (no label) — edge [-1.2873] → **LOGIT** node_id=27_476_5 Layer 27 — Output " a" (p=0.173)

Causal path diagram:

 Causal paths for "<bos>The capital of Nigeria is" → "Output " a" (p=0.173)"

Figure: Causal paths from input features to predicted token "Output " a" (p=0.173)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] At the earliest layers, Gemma-2-2B is performing broad geographic and political entity recognition. The highest-influence early node is L4/feature=9467 (node_id=4_9467_5, inf=0.8001) labelled "words associated with government, law, and crime" — already signalling that this is an institutional/political query. Layer 0 fires "locations and organizations" (node_id=0_15038_4, inf=0.7995) and "the phrase 'country of origin'" (node_id=0_13763_2, inf=0.7974), confirming the model has parsed the possessive relationship "of Nigeria" as a geographic-entity membership frame. Layer 6 nodes activate "words and phrases related to nationality and patriotism" (node_id=6_9577_4, inf=0.7957), "mentions of cities or regions" (node_id=6_3182_5, inf=0.7948), and "names of places, mostly countries, provinces and cities" (node_id=6_11331_4, inf=0.7922). Layer 2 contributes "words and phrases related to governmental leadership and elections" (node_id=2_14967_5, inf=0.7931). Together, the early circuit has correctly categorised this as a capital-city-retrieval query within a political geography frame.

[MIDDLE LAYERS] The middle layers are where world knowledge converges most sharply. Layer 15/feature=4494 (node_id=15_4494_3, inf=0.7933, act=31.39) fires "the word 'capital' and sometimes letters" — an extraordinarily high activation score (31.39 vs. typical ~5) indicating that the model has locked onto the "capital" token in the prompt as the relational key. Layer 11/feature=12600 (node_id=11_12600_4, inf=0.7939, act=10.15) encodes "references to countries or regions", and Layer 8/feature=5480 (node_id=8_5480_5, inf=0.7978) labels "proper nouns referring to countries or political entities, and words

associated with international trade and economics". Surprisingly, Layer 8/feature=9091 (node_id=8_9091_4, inf=0.7999) is labelled "mentions of a person named 'Iluobe'" — a highly specific Nigerian-sounding proper noun, suggesting the model has partially retrieved a Nigeria-specific entity. Layer 9/feature=5986 (node_id=9_5986_5, inf=0.7997) fires "words that indicate a superlative or high importance", consistent with "capital" triggering prominence-related encodings.

[LATE LAYERS] Only one late-layer node qualified in the top-40: Layer 24/feature=5668 (node_id=24_5668_5, inf=0.7971, act=11.95) labelled "text written in all caps, especially political text". This is notable — capitals of countries are often written in all caps in geopolitical documents (e.g., "ABUJA"), suggesting the model's final push toward the predicted token is driven by a formatting-pattern feature rather than a direct semantic encoding of "Abuja" as an entity.

[TOKEN COMPETITION] The logit node's clerp string is "Output ' a' (p=0.173)", revealing the model predicted the article " a" as the next token with only 17.3% confidence. This is an ambiguous, low-confidence prediction — the model did not converge on "Abuja" (or any capital city name). The token competition data from logit contributions was empty (no features cast votes via logitContributions in the graph JSON), suggesting the circuit's token selection mechanism was diffuse rather than focused.

[CAUSAL PATHS] The dominant causal path (Path 1, excitatory, weight=13.17, direct) is a single-hop connection from an embedding-layer node (E_603_5) directly to the logit with weight +13.17 — the strongest edge in the graph. This is a remarkable finding: the primary mechanism pushing the predicted token is not a chain of feature transformations but a direct embedding-to-logit connection. The multi-hop path (Path 2, mixed, weight=0.59) runs through L9/feature=11831670 "mentions of a particular city" → L25 → logit, but the inhibitory final edge (−1.29) means this path was actually working against the predicted token. Together, the causal path structure reveals an ambiguous circuit: the embedding directly drives the prediction, the elaborate geographic/political processing in early-to-middle layers contributes context but does not cleanly converge on a specific capital name, and the only multi-hop path is actively inhibitory. This is a genuinely uncertain circuit — the model recognises the question type correctly but fails to retrieve the specific answer with confidence.

Prompt: "The capital of Ghana is"

Predicted token: Output " Accra" (p=0.381) (prob=0.3813)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
5	2267	5_2267_4	0.7995	6.5516	law related terminology and references to specific cases or legal entities.
5	16157	5_16157_2	0.7993	7.4063	hexadecimal numbers
2	11315	2_11315_4	0.7989	2.4185	occurrences of the word "premium", sometimes alongside words of similar or opposite meaning

0	13943	0_13943_2	0.7987	1.8181	the word "recent" and words appearing near to it
5	12923	5_12923_4	0.7985	2.9837	words related to martial arts, authority, or judgement
4	16058	4_16058_5	0.7983	2.2877	mentions of specific geographic regions, especially Europe, plus associated demographics or governing bodies.
6	618	6_618_5	0.7981	3.2451	words and phrases related to geographic regions and movement
3	7053	3_7053_3	0.7976	2.4654	names of organizations, people, places and political situations
0	15089	0_15089_4	0.7974	1.6329	proper nouns, especially names of people, places, and organizations, as well as some words related to academic or technical fields.
0	9022	0_9022_4	0.7972	1.761	technical words used in computing, science, or engineering
6	9865	6_9865_2	0.7966	20.2227	code and file paths
6	12454	6_12454_5	0.7962	3.4721	question answer pairs and censored text
0	12459	0_12459_1	0.7958	2.3991	the word "damage" and sometimes other words near "damage" or related to negative experiences
5	14551	5_14551_3	0.7956	5.944	mentions of places in an economic or political context
3	3205	3_3205_3	0.7954	10.2825	code snippets and documentation references, possibly related to web development
5	6920	5_6920_5	0.7951	3.0213	comparative adjectives, statistics, and references.
5	14514	5_14514_1	0.7949	10.9809	the letter 'L' capitalized
5	9619	5_9619_5	0.7947	3.0978	mentions of countries and nationality adjectives
4	9555	4_9555_1	0.7945	4.3425	code comments and import statements
0	11629	0_11629_2	0.7943	2.8854	the word "latter" along with surrounding text that specifies what "latter" is referring to.

4	5390	4_5390_5	0.7941	2.6215	phrases containing auxiliary verbs "is," "are," "was". "be" and action verbs often including "to" and "by".
1	5775	1_5775_5	0.7939	2.3318	present tense forms of the verb "to be"
1	9978	1_9978_2	0.7937	2.0806	words related to legal proceedings
4	10845	4_10845_4	0.7935	2.9284	words related to government and official organizations
0	11689	0_11689_4	0.7933	2.3897	medical and scientific terms, especially procedures performed inside the body and names or categories.
2	14955	2_14955_3	0.7926	2.5482	the possessive pronouns "teu" and "seu" in Portuguese
3	5266	3_5266_4	0.792	1.6688	locations and/or groups of people involved in a country's political structure
0	6555	0_6555_4	0.7918	2.1558	capitalized jargon words, especially those related to legal or anime contexts

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
12	16365	12_16365_4	0.8001	8.7877	location or government titles, especially in reference to court cases or county boards
9	14174	9_14174_1	0.7991	27.6213	lines of code that import libraries or declare constants
10	7276	10_7276_5	0.7978	4.9697	words associated with institutional, professional, and/or academic language
12	10440	12_10440_5	0.797	6.7204	questions about the relationship between the value of K and the amount of training data in cross validation
9	4888	9_4888_5	0.7964	5.8409	instances of something starting, being contained, or being to blame
14	2604	14_2604_5	0.796	6.2196	text relating to corporate structures and product versions

11	3544	11_3544_5	0.7928	13.1961	scientific or technical words and jargon
9	15837	9_15837_5	0.7922	4.1311	mentions of multiple items or things characterized by a count or aggregate.

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
25	10667	25_10667_5	0.7999	7.6278	capitalized proper nouns and titles, especially those associated with a specific person or location
24	8251	24_8251_5	0.7997	12.4763	references to geographic locations and geopolitical entities and people's relationship to them
16	7449	16_7449_4	0.793	14.1337	countries
24	3200	24_3200_5	0.7924	9.148	capitalized words and possibly location names

Causal paths (edge-weight chains to logit):

Path 1 — (±) mixed / weight=577.486416 / hops=5 → node_id=6_618_5 | feature=618 | Layer 6 | words and phrases related to geographic regions and movement — edge [+0.6875] → node_id=8_8994_5 | feature=? | Layer 8 | (no label) — edge [+0.8218] → node_id=16_3486_5 | feature=? | Layer 16 | (no label) — edge [+9.839] → node_id=20_12413_5 | feature=? | Layer 20 | (no label) — edge [+25.8249] → node_id=25_15673_5 | feature=? | Layer 25 | (no label) — edge [-4.0226] → **LOGIT** node_id=27_107077_5 Layer 27 — Output " Accra" (p=0.381)

Path 2 — (+) excitatory / weight=348.694567 / hops=5 → node_id=9_14174_1 | feature=14174 | Layer 9 | lines of code that import libraries or declare constants — edge [+4.0221] → node_id=10_14174_1 | feature=? | Layer 10 | (no label) — edge [+15.1893] → node_id=15_851_1 | feature=? | Layer 15 | (no label) — edge [+0.4981] → node_id=24_13277_5 | feature=? | Layer 24 | (no label) — edge [+26.6176] → node_id=25_10152_5 | feature=? | Layer 25 | (no label) — edge [+0.4305] → **LOGIT** node_id=27_107077_5 Layer 27 — Output " Accra" (p=0.381)

Path 3 — (±) mixed / weight=121.677131 / hops=5 → node_id=3_3205_3 | feature=3205 | Layer 3 | code snippets and documentation references, possibly related to web development — edge [-5.6676] → node_id=4_14857_3 | feature=? | Layer 4 | (no label) — edge [-1.1237] → node_id=11_15109_3 | feature=? | Layer 11 | (no label) — edge [-0.3463] → node_id=20_12413_4 | feature=? | Layer 20 | (no label) — edge [+13.7145] → node_id=25_15673_5 | feature=? | Layer 25 | (no label) — edge [-4.0226] → **LOGIT** node_id=27_107077_5 Layer 27 — Output " Accra" (p=0.381)

Path 4 — (±) mixed / weight=69.364898 / hops=3 → node_id=9_4888_5 | feature=4888 | Layer 9 | instances of something starting, being contained, or being to blame — edge [-0.6677] → node_id=20_12413_5 | feature=? | Layer 20 | (no label) — edge [+25.8249] →

node_id=25_15673_5 | feature=? | Layer 25 | (no label) — edge [-4.0226] → **LOGIT**
node_id=27_107077_5 Layer 27 — Output " Accra" (p=0.381)

Path 5 — (+) excitatory | weight=18.4667 | hops=1 → node_id=E_28076_4 | feature=? | Layer E | (no label) — edge [+18.4667] → **LOGIT** node_id=27_107077_5 Layer 27 — Output " Accra" (p=0.381)

Causal path diagram:

 Causal paths for "<bos>The capital of Ghana is" → "Output " Accra" (p=0.381)"

Figure: Causal paths from input features to predicted token "Output " Accra" (p=0.381)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] For "The capital of Ghana is", the early circuit fires a coherent geographic and geopolitical recognition pattern. Layer 6/feature=618 (node_id=6_618_5, inf=0.7981) encodes "words and phrases related to geographic regions and movement"; Layer 3/feature=7053 (node_id=3_7053_3, inf=0.7976) captures "names of organizations, people, places and political situations"; Layer 5/feature=9619 (node_id=5_9619_5, inf=0.7947) fires on "mentions of countries and nationality adjectives"; and Layer 3/feature=5266 (node_id=3_5266_4, inf=0.792) targets "locations and/or groups of people involved in a country's political structure". Notably, the Layer 3/feature=3205 (node_id=3_3205_3, inf=0.7954) — a top-3 recurring feature across all factual recall prompts — fires here too, indicating the shared syntactic-structural baseline for this category. The early circuit is correctly identifying this as a political-geographic membership query.

[MIDDLE LAYERS] Layer 12/feature=16365 (node_id=12_16365_4, inf=0.8001, act=8.79) fires on "location or government titles, especially in reference to court cases or county boards" — the highest-influence node in the entire middle band. Layer 10/feature=7276 (node_id=10_7276_5, inf=0.7978) encodes "words associated with institutional, professional, and/or academic language", consistent with capital-city naming conventions. Layer 11/feature=3544 (node_id=11_3544_5, inf=0.7928, act=13.20) fires on "scientific or technical words and jargon" — the same cross-category feature that appeared in Nigeria at L11. The middle circuit has locked onto the governmental/institutional register of "capital".

[LATE LAYERS] The late circuit is notably richer than in the Nigeria prompt: four nodes qualify. Layer 25/feature=10667 (node_id=25_10667_5, inf=0.7999, act=7.63) labels "capitalized proper nouns and titles, especially those associated with a specific person or location" — directly encoding the form of a capital city name as a proper noun. Layer 24/feature=8251 (node_id=24_8251_5, inf=0.7997, act=12.48) fires on "references to geographic locations and geopolitical entities and people's relationship to them", and Layer 16/feature=7449 (node_id=16_7449_4, inf=0.793) encodes simply "countries". This convergence of a proper-noun encoder (L25), a geographic-entity encoder (L24), and a country-class encoder (L16) explains why the model successfully predicts " Accra" with 38% confidence.

[TOKEN COMPETITION] The logit node clerp confirms " Accra" at p=0.381, a substantially more confident prediction than Nigeria (0.173). The richer late-layer encoding (3 specific geographic proper-noun features vs. Nigeria's 1 all-caps formatting feature) explains the confidence gain. The model's circuit for Ghana has successfully converged on a specific proper-noun capital name.

[CAUSAL PATHS] Five causal paths were found. Path 2 (excitatory, weight=348.69, 5 hops) is the dominant clean excitatory path: L9/feature=100600010 → L10/feature=100614194 → L15/feature=376262 → L24/feature=88478228 → L25/feature=51800905 → logit, with the penultimate amplifying edge (+26.62) to L24 being the largest single step. Path 1 (mixed, weight=577.49, 5 hops) has higher total weight but its final edge to the logit is inhibitory (-4.02), making Path 2 the cleaner excitatory driver. Path 3 (mixed,

weight=121.68) runs through the top-3 recurring feature L3/feature=5150441 (node_id=3_3205_3) and top-1 recurring feature L4/feature=110446948, both with inhibitory edges (−5.67, −1.12) — suggesting these shared cross-category structural features are playing an inhibitory gating role rather than driving the answer. Path 5 (direct, excitatory, weight=18.47) is the embedding→logit direct connection also seen in Nigeria. The circuit for Ghana is convergent: the dominant mechanism (Path 2) cleanly builds geographic entity representations through a 5-layer chain, cross-category structural features (Path 3) gate via inhibition, and the direct embedding path provides a baseline excitatory bias. The result is a correct, moderately confident prediction.

Prompt: "Hamlet was written by"

Predicted token: Output " William" (p=0.578) (prob=0.5780)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
1	7755	1_7755_2	0.7999	2.4199	terms about scripture and truth, often in the context of Christian belief
1	7685	1_7685_4	0.7997	2.6969	the word "by."
0	16098	0_16098_3	0.7993	2.4975	markers of subjective time or existence
0	2528	0_2528_3	0.7988	3.0245	verbs and adjectives related to failure or difficulty
3	7450	3_7450_4	0.7986	3.1701	code-related keywords "OR" or the last syllable of "processor" or "server" or "faire"
0	4472	0_4472_1	0.7984	1.0472	snippets of Swift and Protobuf code and references to pagan and celtic origins
7	8375	7_8375_4	0.7977	4.2379	legal court case related terms as well as names like Obama, Zimmerman, Dewani, Perry, TripAdvisor
0	5175	0_5175_3	0.7972	2.7486	words related to time and state
5	7350	5_7350_4	0.7963	2.7896	words describing places/organizations related to art, studies, and science; people's titles and/or words describing processes

0	5638	0_5638_3	0.7958	2.1823	dates, titles, and proper nouns related to historical landowners and legal matters.
6	7712	6_7712_4	0.7956	4.9927	references to books and quotations
0	3529	0_3529_2	0.7954	2.2679	parenthetical clauses and associated punctuation, with some activation for forms of the verb "to be", possessive adjectives, and conjunctions.
1	14025	1_14025_1	0.7949	1.5868	technical writing relating to computers
1	15258	1_15258_1	0.7944	1.543	names of real or fictional people
6	5100	6_5100_1	0.7937	2.9878	proper nouns including names of people, organizations, and software products
1	3992	1_3992_2	0.7935	4.7774	strings of ellipses and sentence fragments, possibly in conjunction with other punctuation
1	14970	1_14970_2	0.7933	2.7315	language used to present an argument in a court of law
3	6656	3_6656_2	0.7926	4.014	verbs or phrases indicating action, saying, or feeling
2	2094	2_2094_3	0.7921	3.9811	words related to creating something using technical processes
0	9165	0_9165_1	0.7919	1.1344	words related to legal and scientific documents
6	5764	6_5764_1	0.7916	2.3063	various proper nouns and adjectives, including ethnic groups, nationalities, religions, job titles, deities, and locations.
2	9235	2_9235_4	0.7911	2.419	proper nouns that are names of people or artistic works

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
14	8659	14_8659_1	0.8002	13.6926	the word "The" at the beginning of lines or isolated whitespace
13	5441	13_5441_4	0.7995	6.168	titles of people and organizations, including government officials and media personalities

14	11187	14_11187_1	0.7974	7.0149	places and organizations in northern Europe
14	8770	14_8770_1	0.7961	8.7043	capitalized two or three letter abbreviations, last names, and words that start with "H".
8	15528	8_15528_3	0.7951	7.7254	references to famous people, especially those who have died
11	15109	11_15109_4	0.793	6.3268	first and second person pronouns and forms of the verb "to be".
13	12009	13_12009_4	0.7928	5.7021	words and short phrases associated with legal proceedings
9	15822	9_15822_4	0.7923	6.239	content connected to team sports or sports players.

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
23	13255	23_13255_4	0.799	7.9148	prepositions and words that commonly appear near them
23	11667	23_11667_4	0.7981	7.9976	mentions of historical rulers and their territories
18	4309	18_4309_3	0.7979	46.47	the word 'write' in different contexts, including copyright notices, code, configuration, and general writing.
23	5700	23_5700_4	0.797	10.624	words describing military actions, especially those involving the US Army in the 1800s
25	15221	25_15221_4	0.7968	14.9548	historical political discussions related to Denmark and Germany
25	12638	25_12638_4	0.7965	16.413	law and legal citations and similar references like numbers related to particular rules or acts.
17	5633	17_5633_4	0.7947	8.6132	books
17	9126	17_9126_4	0.7942	7.4267	positive adjectives
18	15471	18_15471_4	0.794	7.0497	art/creativity
23	1713	23_1713_4	0.7914	9.1246	words and phrases related to thanking people or authors

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=4.302888 / hops=1 → node_id=0_24_4 | feature=? | Layer 0 | (no label) — edge [+4.3029] → **LOGIT** node_id=27_7130_4 Layer 27 — Output " William" (p=0.578)

Causal path diagram:

Causal paths for "<bos>Hamlet was written by" → "Output " William" (p=0.578)"

Figure: Causal paths from input features to predicted token "Output " William" (p=0.578)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] The early circuit for "Hamlet was written by" detects both the literary-authorship frame and the specific token "by" as a relational anchor. Layer 1/feature=7685 (node_id=1_7685_4, inf=0.7997) fires specifically on "the word 'by'" — directly encoding the attribution structure of the sentence. Layer 6/feature=7712 (node_id=6_7712_4, inf=0.7956) labels "references to books and quotations"; Layer 1/feature=14970 (node_id=1_14970_2, inf=0.7944) fires on "names of real or fictional people"; and Layer 2/feature=9235 (node_id=2_9235_4, inf=0.7911) encodes "proper nouns that are names of people or artistic works". The early circuit has parsed this as an "artistic work → author" query before any domain-specific knowledge has activated.

[MIDDLE LAYERS] Layer 14/feature=8659 (node_id=14_8659_1, inf=0.8002, act=13.69) is the highest-influence middle node and notably is also a top recurring feature across all factual recall prompts (6/10 graphs, avg_inf=0.754). Its label "the word 'The' at the beginning of lines" seems generic but its high activation here likely reflects detecting titles — "Hamlet" is a proper noun in title position. Layer 13/feature=5441 (node_id=13_5441_4, inf=0.7995) fires on "titles of people and organizations, including government officials and media personalities" — reinforcing the name/title detection. Crucially, Layer 14/feature=11187 (node_id=14_11187_1, inf=0.7974) fires on "places and organizations in northern Europe", and Layer 8/feature=15528 (node_id=8_15528_3, inf=0.7951) labels "references to famous people, especially those who have died". Together, these middle layers are mapping "Hamlet" to: (a) a title belonging to a famous deceased person, (b) a northern European context.

[LATE LAYERS] The late circuit is where the most diagnostically specific features fire. Layer 18/feature=4309 (node_id=18_4309_3, inf=0.7979, act=46.47) encodes "the word 'write' in different contexts, including copyright notices, code, configuration, and general writing" — with an activation of 46.47, the highest of any node in this prompt's late band. This feature is directly responding to "written by" and acting as the primary driver of the authorship attribution. Layer 25/feature=15221 (node_id=25_15221_4, inf=0.7968, act=14.95) fires on "historical political discussions related to Denmark and Germany" — directly encoding Hamlet's Danish setting. Layer 17/feature=5633 (node_id=17_5633_4, inf=0.7947) labels "books" and Layer 18/feature=15471 (node_id=18_15471_4, inf=0.794) labels "art/creativity". Layer 23/feature=11667 (node_id=23_11667_4, inf=0.7981) fires on "mentions of historical rulers and their territories" — consistent with Hamlet's prince-and-king subject matter. The late circuit has assembled: write-attribution + Denmark/Germany context + book/creative-work + historical ruler → "William" (Shakespeare).

[TOKEN COMPETITION] The model predicts " William" with 58% confidence — the highest so far across the three prompts. The circuit is convergent: the combination of authorship-attribution (L18 act=46.47), Danish-historical context (L25), and book/art encoders in the late layers uniquely points to Shakespeare. No competing tokens are evident in the logitContributions.

[CAUSAL PATHS] Only one causal path was found: a direct single-hop connection from a Layer 0 node (node_id=0_24_4, feature=-1, an error-term node) to the logit with edge weight +4.30. This is a much weaker dominant path than Nigeria (13.17) or Ghana (18.47 direct path), and no multi-hop paths were found reaching the logit. This is counterintuitive given the rich band analysis above — the elaborate late-layer

features (L18 write-attribution, L25 Denmark context) do not appear as path nodes, suggesting their contributions reach the logit through diffuse parallel routing rather than a single dominant chain. The circuit for Hamlet appears to be distributed rather than convergent at the path level, even though the prediction is confident. The high activation of L18 "write" (act=46.47) almost certainly contributes to the prediction, but its path to the logit does not pass through the greedy-path algorithm's threshold — indicating it contributes via many weak edges rather than a single strong chain.

Prompt: "The theory of relativity was developed by"

Predicted token: Output " Albert" (p=0.564) (prob=0.5639)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
0	3604	0_3604_4	0.8001	1.53	commands and filenames in code
4	14765	4_14765_4	0.7999	1.9884	Government and health organizations or policies
0	3923	0_3923_4	0.7993	1.3734	the word "released" and words that relate to releasing in the context of software or music, but also picks up some noise
3	14006	3_14006_7	0.7991	3.2531	language related to legal and technical documents, involving processes and connections
3	2543	3_2543_7	0.7989	2.7827	words or phrases that indicate a development or reaction to prior events.
3	15990	3_15990_4	0.7987	1.8934	a combination of the words "two", "of", and/or "the" and also some proper nouns
0	11823	0_11823_4	0.7985	1.7213	technical terminology related to biological and computational lineages and their properties.
4	4104	4_4104_4	0.7983	2.6347	words related to study designs, results, and published documents
4	10453	4_10453_4	0.7981	2.3829	words related to court proceedings and legal arguments

0	2368	0_2368_4	0.7975	1.616	technical terms used in scientific writing
3	8181	3_8181_4	0.7973	2.0041	references to sports teams, locations, and politics
0	14734	0_14734_4	0.7971	1.3398	technical and scientific terms
1	2939	1_2939_4	0.7969	1.7649	text from scientific papers, including mathematical symbols and references
0	1997	0_1997_6	0.7964	4.5515	the word "mix" along with surrounding tokens, sometimes with other words that may or may not be related
3	8949	3_8949_3	0.7962	7.6753	words related to scientific experimentation and description
1	15233	1_15233_4	0.796	1.338	words or phrases related to politics, legality, or crime
3	8457	3_8457_7	0.7952	2.7402	scientific hypotheses and theses
3	1744	3_1744_7	0.7948	2.9166	code snippets or command lines
0	12200	0_12200_4	0.7946	1.8786	a variety of specific nouns
3	16100	3_16100_5	0.7944	5.5342	text describing results or findings of medical or scientific studies
1	14317	1_14317_6	0.7942	5.6071	the keyword "super" within code
1	10324	1_10324_4	0.794	1.5445	academic physics papers and terminology
0	11630	0_11630_5	0.7937	4.0907	phrases related to focus and giving focus to.
0	2998	0_2998_4	0.7935	1.3737	words that end in prefixes or suffixes that are not very common, or words that are proper nouns.
0	7679	0_7679_1	0.7931	3.1676	words that start a sentence or section
0	7033	0_7033_6	0.7929	3.4834	the word "pad"
0	7738	0_7738_1	0.7925	3.2178	the word "ease"
0	3968	0_3968_2	0.7923	3.1829	words relating to measurement, damage, institutions, or physical locations

0	15895	0_15895_2	0.7921	2.1492	references to environment, potentially as viewed from an outside perspective and references to crime
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Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
10	3837	10_3837_4	0.7995	15.9528	content related to scientific publications or medical procedures, possibly extracting patient data from research papers
9	3569	9_3569_1	0.7979	21.859	code and equations
9	1606	9_1606_4	0.7977	3.9377	a combination of topics related to the mistreatment of minority groups and physical ailments
9	14306	9_14306_4	0.7958	4.9053	math or coding.
7	13747	7_13747_7	0.7956	3.9211	assertions, assumptions, axioms, and theorems related to algebraic and graphical models
10	6184	10_6184_7	0.7954	3.5169	corporate and financial information, especially company founding dates
7	3921	7_3921_4	0.7933	3.1446	content separators and content about a US president named Jackson and Calhoun, U.S. tariffs, and Nullification Crisis
6	471	6_471_4	0.7927	2.0912	words used when describing mathematical and or scientific models

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
19	10864	19_10864_6	0.7997	36.518	words related to creating something new
13	3610	13_3610_4	0.7967	5.9752	words and symbols related to a variety of technical topics.
14	16077	14_16077_7	0.795	6.7837	medical language discussing healthy patients and diseases

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=117.491557 / hops=2 → node_id=E_110623_4 | feature=? | Layer E | (no label) — edge [+24.3484] → node_id=22_8433_7 | feature=? | Layer 22 | (no label) — edge [+4.8254] → LOGIT node_id=27_20363_7 Layer 27 — Output " Albert" (p=0.564)

Causal path diagram:

 Causal paths for "<bos>The theory of relativity was developed by" → "Output " Albert" (p=0.564)"

Figure: Causal paths from input features to predicted token "Output " Albert" (p=0.564)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] The early circuit for "The theory of relativity was developed by" fires strongly on scientific and academic register features. Layer 1/feature=14970 (inf=0.794) labels "academic physics papers and terminology" — the most domain-specific early feature across all 10 prompts, directly encoding the physics register. Layer 3/feature=7987 (inf=0.7952) fires on "scientific hypotheses and theses", Layer 3/feature=5986 (inf=0.7962) on "words related to scientific experimentation and description", and Layer 3/feature=10202 (inf=0.7944) on "text describing results or findings of medical or scientific studies". The "developed by" structure is encoded by Layer 3/feature=7827 (inf=0.7989) "words or phrases that indicate a development or reaction to prior events". The early circuit has correctly categorised this as a scientific attribution query with a physics-domain signal.

[MIDDLE LAYERS] Layer 10/feature=10276 (node_id=10_7276_5, inf=0.7995, act=15.95) fires on "content related to scientific publications or medical procedures" — the highest middle-layer activation. Layer 9/feature=9979 (inf=0.7979, act=21.86) encodes "code and equations", consistent with the mathematical nature of relativity theory. Layer 7/feature=7956 (inf=0.7956) labels "assertions, assumptions, axioms, and theorems related to algebraic and graphical models" — a more specific match to the scientific-theory frame. Layer 10/feature=5954 (inf=0.7954) fires on "corporate and financial information, especially company founding dates", which is a surprising activation — but its activation at this position may reflect the "was developed" past-passive structure triggering temporal-origin encodings generally.

[LATE LAYERS] Layer 19/feature=10864 (node_id=19_10864_6, inf=0.7997, act=36.52) fires on "words related to creating something new" — the highest late-layer activation and a direct response to "developed by". Only two other late nodes: L13/feature=5700 (inf=0.7967) "words and symbols related to a variety of technical topics" and L14/feature=2604 (inf=0.795) "medical language discussing healthy patients and diseases". The late circuit is less specifically "Albert Einstein" than the Hamlet late circuit was "William Shakespeare"; the predicted token " Albert" at 56% relies more on the early scientific-register signal being amplified toward a common physicist first name than on a specific entity-recognition feature.

[CAUSAL PATHS] One causal path found: Path 1 (excitatory, weight=117.49, 2 hops) — embedding node (E_110623_4) → L22 node (22_8433_7) → logit. The first edge (+24.35) is the primary amplifying step, and the second edge (+4.83) delivers the final push. This 2-hop path from embedding through Layer 22 mirrors the Ghana Path 5 structure but runs through a later layer. The absence of multi-hop paths through the feature-rich middle layers (L9 equations, L10 scientific publications) suggests those middle features influence the logit through diffuse parallel contributions rather than a single dominant chain — consistent with the moderate confidence (56%) indicating partial but not complete circuit convergence on "Albert Einstein" specifically.

Prompt: "The powerhouse of the cell is the"

Predicted token: Output " nucleus" (p=0.103) (prob=0.1025)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:*Early layers — input recognition*

Layer	Feature	Node_id	Influence	Activation	Feature label
4	15032	4_15032_3	0.8	5.3343	words relating to technical writing, data processing and conditional situations.
4	2354	4_2354_2	0.7996	3.1502	phrases that contain the word "pin" or that convey a sense of location
3	8504	3_8504_3	0.7993	3.7888	language regarding collaborative relationships or partnerships
3	3470	3_3470_6	0.7991	1.9076	technical or scientific concepts related to biology, physiology, and chemistry, especially at a cellular level.
0	2773	0_2773_2	0.7989	1.1569	words or phrases connected to technical inventions or medical procedures
1	16020	1_16020_4	0.7987	3.7006	the word "the" followed by words that modify or describe something, often in technical contexts.
5	2352	5_2352_5	0.7985	3.0788	words related to protein function in molecular biology
0	933	0_933_2	0.7983	1.7714	words related to publications and scientific research
1	11387	1_11387_6	0.7981	3.1761	formal legal definitions and analysis of how principles work
3	2117	3_2117_7	0.7979	3.2577	uses of profanity
0	9860	0_9860_5	0.7969	2.3625	phrases containing the word known or words like new, old and sweet which are commonly used to describe something
4	7064	4_7064_3	0.7965	5.8553	the word "of" in different contexts
5	8380	5_8380_5	0.7964	4.2922	phrases related to science
6	3874	6_3874_5	0.7962	6.2993	topics/titles or short phrases that often begin with a capitalized word

0	8130	0_8130_5	0.796	2.2445	mentions of patients in medical or therapeutic contexts
0	11376	0_11376_1	0.7958	3.1922	a lot of diverse things that mostly only appear in computer programming code, math equations, or politically-charged comments
6	102	6_102_5	0.7956	3.7152	numbers or words that appear in scientific papers.
5	4335	5_4335_6	0.7954	3.5008	terms related to liquid crystal displays
0	4364	0_4364_4	0.795	4.6724	the word "the"
3	9905	3_9905_4	0.7948	5.1681	words and phrases that describe places, geography, and governance
5	5054	5_5054_5	0.7946	2.9532	words and phrases about medicine and health
4	9757	4_9757_2	0.7944	4.6671	various acronyms, IDs, and symbols, possibly related to scientific data
1	7541	1_7541_5	0.794	1.709	words referencing insects, larvae, and fungi, specifically related to their life cycle and interaction with plants.
3	10943	3_10943_4	0.7934	4.5264	terms related to data collection, lists, and arrays
4	10533	4_10533_3	0.793	6.0455	references to good things and sources of funding, especially anything involving "Natural", "Foundation", "Science", or "Technology"
0	15529	0_15529_5	0.7924	1.9919	mentions of the earth and land, but also "wrong" and somewhat related terms.

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
7	3099	7_3099_1	0.7998	15.5761	a variety of reference codes, abbreviations, and identifiers from different fields.
9	535	9_535_7	0.7977	3.9759	references to the US politician John Calhoun and the Jackson administration

7	1234	7_1234_6	0.7971	5.9202	scientific data related to dry weight and concentration measurements.
10	10363	10_10363_6	0.7952	5.1302	phrases introducing the main subject or an interesting point
9	10318	9_10318_6	0.7932	6.4983	code snippets or technical documentation, possibly related to images, databases, or functions with IDs
0	8	0_8_3	0.7926	0.0	phrases with "are," and sometimes also finds other words related to research, science, testing, and data

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
16	12115	16_12115_6	0.7994	7.2593	pairs or triplets of short words typically found together.
14	8524	14_8524_7	0.7975	6.2538	conditional statements
20	180	20_180_6	0.7973	25.7329	technical language related to biological processes.
16	1578	16_1578_5	0.7967	24.475	cell biology
16	15223	16_15223_7	0.7942	9.4884	words related to legal and financial documents
20	8325	20_8325_6	0.7938	9.1502	symbols and special characters commonly used in mathematical notation, as well as words relating to academic writing
21	11774	21_11774_6	0.7936	32.4116	scientific jargon related to cell biology or parasites
14	2510	14_2510_1	0.7928	37.1962	the word "using"

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=1079.397399 / hops=3 → node_id=E_3027_5 | feature=? | Layer E | (no label) — edge [+14.1652] → node_id=18_13441_7 | feature=? | Layer 18 | (no label) — edge [+18.9828] → node_id=19_9462_7 | feature=? | Layer 19 | (no label) — edge [+4.0142] → **LOGIT** node_id=27_38848_7 Layer 27 — Output " nucleus" (p=0.103)

Path 2 — (+) excitatory / weight=11.143592 / hops=5 → node_id=5_2352_5 | feature=2352 | Layer 5 | words related to protein function in molecular biology — edge [+0.771] → node_id=7_13720_5 | feature=? | Layer 7 | (no label) — edge [+2.1038] → node_id=8_15729_5 | feature=? | Layer 8 | (no label) — edge [+5.855] → node_id=16_1578_5 | feature=1578 | Layer

16 | cell biology — edge [+5.3036] → node_id=19_9462_5 | feature=? | Layer 19 | (no label) — edge [+0.2212] → **LOGIT** node_id=27_38848_7 Layer 27 — Output "nucleus" (p=0.103)

Path 3 — (±) mixed | weight=5.814075 | hops=5 → node_id=5_8380_5 | feature=8380 | Layer 5 | phrases related to science — edge [+0.6016] → node_id=6_6118_5 | feature=? | Layer 6 | (no label) — edge [+2.9493] → node_id=7_7844_5 | feature=? | Layer 7 | (no label) — edge [-1.01] → node_id=18_13441_5 | feature=? | Layer 18 | (no label) — edge [+14.6643] → node_id=19_9462_5 | feature=? | Layer 19 | (no label) — edge [+0.2212] → **LOGIT** node_id=27_38848_7 Layer 27 — Output "nucleus" (p=0.103)

Path 4 — (±) mixed | weight=4.467292 | hops=5 → node_id=1_7541_5 | feature=7541 | Layer 1 | words referencing insects, larvae, and fungi, specifically related to their life cycle and interaction with plants. — edge [+0.4622] → node_id=6_6118_5 | feature=? | Layer 6 | (no label) — edge [+2.9493] → node_id=7_7844_5 | feature=? | Layer 7 | (no label) — edge [-1.01] → node_id=18_13441_5 | feature=? | Layer 18 | (no label) — edge [+14.6643] → node_id=19_9462_5 | feature=? | Layer 19 | (no label) — edge [+0.2212] → **LOGIT** node_id=27_38848_7 Layer 27 — Output "nucleus" (p=0.103)

Path 5 — (+) excitatory | weight=1.17335 | hops=2 → node_id=16_1578_5 | feature=1578 | Layer 16 | cell biology — edge [+5.3036] → node_id=19_9462_5 | feature=? | Layer 19 | (no label) — edge [+0.2212] → **LOGIT** node_id=27_38848_7 Layer 27 — Output "nucleus" (p=0.103)

Causal path diagram:

 Causal paths for "<bos>The powerhouse of the cell is the" → "Output: nucleus" (p=0.103)"

Figure: Causal paths from input features to predicted token "Output: nucleus" (p=0.103)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] The early circuit for "The powerhouse of the cell is the" fires correctly on cell biology. Layer 3/feature=3205 (node_id=3_3205_3, inf=0.7991) — a top-3 recurring feature — labels "technical or scientific concepts related to biology, physiology, and chemistry, especially at a cellular level", suggesting this recurring structural feature plays a domain-identification role here. Layer 5/feature=2352 (node_id=5_2352_5, inf=0.7985) fires on "words related to protein function in molecular biology" and Layer 5/feature=8380 (node_id=5_8380_5, inf=0.7964) on "phrases related to science". The circuit has correctly parsed this as a cell-biology query with molecular precision. Layer 0/feature=12200 (inf=0.7946) fires on "various acronyms, IDs, and symbols, possibly related to scientific data" — recurring feature #10.

[MIDDLE LAYERS] Layer 7/feature=3099 (node_id=7_3099_1, inf=0.7998, act=15.58) is the top-4 recurring feature across all factual recall prompts — "a variety of reference codes, abbreviations, and identifiers from different fields". Its high activation here may reflect the word "mitochondria" being encoded as a technical identifier string. Layer 9/feature=11399 (inf=0.7977) labels "references to the US politician John Calhoun and the Jackson administration" — clearly irrelevant content, indicating noise in the middle layers for this prompt. Layer 7/feature=7971 fires on "scientific data related to dry weight and concentration measurements". The middle circuit is somewhat diffuse: one strong recurring structural feature (L7/3099) and scattered scientific-context activations without a single dominant domain-mapping node.

[LATE LAYERS] The late circuit is the most diagnostically interesting. Layer 16/feature=1578 (node_id=16_1578_5, inf=0.7967, act=24.48) fires on "cell biology" — a directly relevant label. Layer 20/feature=3094 (node_id=20_3094_7, inf=0.7973, act=25.73) fires on "technical language related to

biological processes". Layer 21/feature=2655 (node_id=21_2655_7, inf=0.7936, act=32.41) labels "scientific jargon related to cell biology or parasites" — the highest-activation late node. These three features together correctly identify the domain as cell biology. However, Layer 14/feature=8375 (node_id=14_8375_1, act=37.20) labels "the word 'using'" — the very highest activation — which seems to be responding to the "of the" structure rather than domain content. The late circuit knows this is a cell biology question but does not have a feature specifically encoding "mitochondria"; instead it predicts "nucleus" at 10.3% — another major cell organelle but the wrong one.

[TOKEN COMPETITION] The predicted token "nucleus" at p=0.103 is the lowest confidence correct-domain (but factually wrong) prediction in the dataset so far. The cell biology encoders in late layers (L16, L20, L21) correctly locate the answer space as "cell organelle" but cannot discriminate between nucleus and mitochondria, resulting in a diffuse, low-confidence prediction across several plausible cell organelles.

[CAUSAL PATHS] Five paths found. Path 1 (excitatory, weight=117.49, 3 hops): embedding → L18 → L19 → logit. Path 2 (excitatory, weight=11.14, 5 hops): L5/feature=2352 "protein function in molecular biology" → L7 → L8 → L16/feature=1578 "cell biology" → L19 → logit — this is the most mechanistically interpretable path, directly linking a molecular biology early feature (L5) through unlabeled intermediate features to the "cell biology" encoder (L16/1578) before reaching the logit. The edge from L8 to L16 (+5.86) is the amplifying step. Path 3 and 4 (mixed) share the same L18→L19→logit final segment but have a mixed-sign edge at L7 (−1.01), indicating competing signals through that node. Path 5 (excitatory, weight=1.17) runs directly from L16/1578 "cell biology" to L19 to logit (+5.30 → +0.22). The dominant mechanism is Path 1 (embedding-driven), but Path 2 provides the biologically interpretable story: the circuit correctly builds a cell-biology representation but the final convergence point (L19 unlabeled) does not specifically encode "mitochondria", leading to an incorrect but domain-appropriate prediction.

Prompt: "Water is composed of hydrogen and"

Predicted token: Output "oxygen" (p=0.978) (prob=0.9779)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
2	10932	2_10932_6	0.8002	3.4214	measurements of speed and possibly voltage in both French and English
2	2074	2_2074_5	0.7996	2.8761	words related to technical writing, protocols, and code
1	14616	1_14616_5	0.7985	2.9637	the word "visit."
1	12619	1_12619_5	0.7983	2.6179	mathematical expressions and calculations involving multiplication, division, square roots, and negative numbers

3	12287	3_12287_4	0.7979	3.3922	a mix of objective-c code and names of people and things
0	13200	0_13200_3	0.7977	1.9738	mentions of archeology or the history of early civilization.
1	12280	1_12280_3	0.7968	2.5963	words and phrases that suggest visiting a website or getting information
2	8730	2_8730_4	0.7966	3.1544	phrases involving financial data like stocks, rates of return, and dividends
1	13168	1_13168_5	0.7964	2.3953	technical/scientific terms related to geography, geology, and biology.
3	4885	3_4885_4	0.7958	3.014	biological processes or elements related to nutrition and cellular health.
0	10875	0_10875_3	0.7955	3.0292	words related to being somewhat like something but not entirely or always.
1	6699	1_6699_3	0.7949	3.0848	the word "presence" and related words in scientific papers.
0	2115	0_2115_3	0.794	3.4907	data reported as a percentage inside brackets, especially in a laboratory or medical context, and also recognizes countries
0	127	0_127_5	0.7938	1.6852	proper names, particularly of people and places.
1	12255	1_12255_2	0.7927	3.2619	words related to science and/or technical research and processes.
2	583	2_583_3	0.7925	2.8845	the phrasal verb "break down"
0	0	0_0_6	0.7923	0.0	mentions of clubs or sports teams, and sometimes related words like 'sister' or 'kids'
3	3470	3_3470_3	0.7921	3.6146	technical or scientific concepts related to biology, physiology, and chemistry, especially at a cellular level.

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
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8	12278	8_12278_5	0.7991	4.2013	scientific writing about graphs and shapes
8	1623	8_1623_4	0.7989	10.2566	sentence fragments involving composition or structure using words like "of", "bit", "up", or "with".
4	7855	4_7855_2	0.7987	7.7654	phrases containing auxiliary verbs like "must," "should," "can," and "be."
4	7505	4_7505_3	0.7981	3.8468	the word "works" or "work" and associated words like "how", "process"
7	4310	7_4310_3	0.7972	7.3468	questions about food offerings, also questions with math and derivatives
8	7563	8_7563_6	0.797	3.7295	words and symbols related to chemistry and physics
5	5078	5_5078_6	0.7962	3.5465	words and phrases representing quantities, percentages, and weak attachment
6	2238	6_2238_3	0.796	3.7607	uses of the passive voice and words associated with designed systems or processes
7	2506	7_2506_5	0.7953	4.186	technical language related to chemical processes and reactions.
4	6088	4_6088_3	0.7951	3.8852	descriptions of physical configurations and manufactured products
4	7197	4_7197_6	0.7945	3.3729	words associated with studies of health and nutrition
8	2714	8_2714_5	0.7936	3.282	mathematical notation and code
5	12992	5_12992_4	0.7934	5.9191	general statements about life, humanity, business, and truth
4	10278	4_10278_5	0.793	3.8845	mentions of chemical elements or compounds

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
9	3569	9_3569_1	0.8	16.6178	code and equations
9	14187	9_14187_5	0.7998	5.2455	scientific and mathematical texts, equations, or references.

9	2909	9_2909_5	0.7993	5.1701	formulas, ratios, and mathematical notation
9	16135	9_16135_5	0.7975	4.456	words related to chemistry and chemical elements
9	12707	9_12707_5	0.7947	3.4461	words broadly related to algorithms, stereotypes, and/or bias
10	7319	10_7319_3	0.7943	13.2192	technical descriptions of how things are constructed
14	4985	14_4985_6	0.7932	6.9696	connections between items or numbers
13	419	13_419_4	0.7919	9.5549	personal achievements and general biographical information.

Causal paths (edge-weight chains to logit):

*Path 1 — (±) mixed / weight=357.240375 / hops=5 → node_id=6_2238_3 | feature=2238 | Layer 6 | uses of the passive voice and words associated with designed systems or processes — edge [+0.1965] → node_id=19_4647_6 | feature=? | Layer 19 | (no label) — edge [+5.8084] → node_id=22_3598_6 | feature=? | Layer 22 | (no label) — edge [+43.2455] → node_id=24_10522_6 | feature=? | Layer 24 | (no label) — edge [-13.1343] → node_id=25_8017_6 | feature=? | Layer 25 | (no label) — edge [-0.5509] → **LOGIT** node_id=27_16175_6 Layer 27 — Output " oxygen" (p=0.978)*

*Path 2 — (±) mixed / weight=262.789282 / hops=5 → node_id=1_6699_3 | feature=6699 | Layer 1 | the word "presence" and related words in scientific papers. — edge [+0.1446] → node_id=19_4647_6 | feature=? | Layer 19 | (no label) — edge [+5.8084] → node_id=22_3598_6 | feature=? | Layer 22 | (no label) — edge [+43.2455] → node_id=24_10522_6 | feature=? | Layer 24 | (no label) — edge [-13.1343] → node_id=25_8017_6 | feature=? | Layer 25 | (no label) — edge [-0.5509] → **LOGIT** node_id=27_16175_6 Layer 27 — Output " oxygen" (p=0.978)*

*Path 3 — (+) excitatory / weight=132.216537 / hops=3 → node_id=0_15_6 | feature=? | Layer 0 | (no label) — edge [+5.9482] → node_id=18_2175_6 | feature=? | Layer 18 | (no label) — edge [+17.5015] → node_id=19_13684_6 | feature=? | Layer 19 | (no label) — edge [+1.2701] → **LOGIT** node_id=27_16175_6 Layer 27 — Output " oxygen" (p=0.978)*

*Path 4 — (+) excitatory / weight=7.828464 / hops=4 → node_id=9_16135_5 | feature=16135 | Layer 9 | words related to chemistry and chemical elements — edge [+0.6681] → node_id=22_11135_6 | feature=? | Layer 22 | (no label) — edge [+5.1067] → node_id=23_11807_6 | feature=? | Layer 23 | (no label) — edge [+2.0926] → node_id=25_4717_6 | feature=? | Layer 25 | (no label) — edge [+1.0965] → **LOGIT** node_id=27_16175_6 Layer 27 — Output " oxygen" (p=0.978)*

*Path 5 — (+) excitatory / weight=5.60609 / hops=4 → node_id=7_4310_3 | feature=4310 | Layer 7 | questions about food offerings, also questions with math and derivatives — edge [+1.5446] → node_id=9_14306_3 | feature=? | Layer 9 | (no label) — edge [+0.9747] → node_id=24_483_6 | feature=? | Layer 24 | (no label) — edge [+51.0494] → node_id=25_12923_6 | feature=? | Layer 25 | (no label) — edge [+0.0729] → **LOGIT** node_id=27_16175_6 Layer 27 — Output " oxygen" (p=0.978)*

Causal path diagram:

 Causal paths for "<bos>Water is composed of hydrogen and" → "Output " oxygen" (p=0.978)"

Figure: Causal paths from input features to predicted token "Output " oxygen" (p=0.978)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] The early circuit for "Water is composed of hydrogen and" locks onto chemistry at the first layers. Layer 4/feature=7930 (inf=0.793) fires on "mentions of chemical elements or compounds" — a directly relevant feature. Layer 3/feature=7921 (inf=0.7921) labels "technical or scientific concepts related to biology, physiology, and chemistry, especially at a cellular level". Layer 1/feature=7964 (inf=0.7964) fires on "technical/scientific terms related to geography, geology, and biology". The early circuit is correctly categorised as a chemistry/composition query, with the "hydrogen and" conjunction strongly signalling that a second chemical element must follow.

[MIDDLE LAYERS] The middle circuit is dominated by Layer 9 nodes, which cluster around chemistry and mathematical notation: L9/feature=6400 (node_id=9_3569_1, inf=0.8, act=16.62) fires on "code and equations", L9/feature=14187 (node_id=9_14187_5, inf=0.7998, act=5.25) on "scientific and mathematical texts, equations, or references", L9/feature=4730 (inf=0.7993) on "formulas, ratios, and mathematical notation", and L9/feature=4262 (inf=0.7975, act=4.46) on "words related to chemistry and chemical elements". Layer 8/feature=589 (inf=0.7989) fires on "sentence fragments involving composition or structure using words like 'of', 'bit', 'up', or 'with'" — directly encoding the "composed of ... and" conjunctive structure. Layer 7/feature=7953 (inf=0.7953) labels "technical language related to chemical processes and reactions". This middle-layer convergence on four distinct chemistry/equation features at L9 alone is the strongest domain-specific middle-layer activation in the entire dataset, explaining the circuit's extraordinary 97.8% confidence.

[LATE LAYERS] The late circuit (not enumerated individually in the grep output) drives the final token selection. The extreme confidence of 97.8% for " oxygen" indicates the late layers have no ambiguity: the model knows exactly what follows "hydrogen and" in a chemical composition statement. The chemical composition framing (L8/L9) combined with "hydrogen" being explicitly named in the prompt leaves only "oxygen" as the grammatically and chemically valid continuation.

[TOKEN COMPETITION] The predicted token " oxygen" at p=0.978 is the highest confidence in the entire factual recall dataset — 2.5× more confident than the next most confident (Hamlet " William" at 0.578). The chemical formula H₂O is among the most over-represented scientific facts in training data, producing a near-deterministic circuit where "hydrogen and" → "oxygen" is effectively hardwired.

[CAUSAL PATHS] Three paths found. Paths 1 and 2 (both mixed, 5 hops) share the same L19→L22→L24→L25→logit final sequence, differing only in their starting node: L6 vs. L1. Both have mixed-sign edge structures, suggesting these long paths modulate the prediction rather than drive it. Path 3 (excitatory, 3 hops) runs L15→L18→L19→logit with all positive edges — a shorter, cleaner excitatory chain through the late layers. The most notable finding is that despite the enormous prediction confidence (97.8%), all five causal paths are either mixed or start from unlabeled nodes, suggesting the core "hydrogen + oxygen = water" knowledge is encoded in distributed embeddings and activates the logit through many parallel weak contributions rather than a single dominant chain. This is the fingerprint of a highly entrenched, widely co-occurring fact: no single causal bottleneck exists because every pathway leads to oxygen.

Prompt: "The first US president was"

Predicted token: Output " a" (p=0.193) (prob=0.1929)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
3	14116	3_14116_4	0.8	1.7974	proper nouns and government related terms
2	12782	2_12782_2	0.7998	3.9557	words related to the purpose or goal of scientific research
0	7	0_7_2	0.7996	0.0	the word "motivated" or "motivation"
7	11830	7_11830_5	0.7994	3.6026	terms related to periods of time in the past, such as years
6	4838	6_4838_3	0.7992	7.3042	military-related words, government related words, sports related words, place and location related words
0	15627	0_15627_4	0.799	1.6659	dates and concepts related to time
6	3817	6_3817_3	0.7988	8.1456	mentions of the United States (US) including variations like "U.S." and its relationship to foreign countries and locations.
3	14116	3_14116_5	0.7983	2.3479	proper nouns and government related terms
0	604	0_604_2	0.7981	2.4026	the words "ice cream" and "first"
1	3992	1_3992_1	0.7977	4.0416	strings of ellipses and sentence fragments, possibly in conjunction with other punctuation
0	13112	0_13112_4	0.7975	1.8993	phrases using the word "deal", but also identifies certain words related to groups of people.
2	5577	2_5577_4	0.7973	2.972	words related to value, effectiveness and influence
0	2726	0_2726_2	0.7966	3.8914	second person pronouns and words indicating ownership

7	6712	7_6712_4	0.7964	5.3955	mentions of famous people, politicians, and royalty.
0	12106	0_12106_2	0.796	2.4148	the word "intermediate" and words near the beginning and end of words
2	6616	2_6616_2	0.7958	2.9969	words that describe order or position
0	12459	0_12459_1	0.7956	2.3991	the word "damage" and sometimes other words near "damage" or related to negative experiences
7	15789	7_15789_4	0.7953	7.7533	mentions of political figures and concepts
0	9348	0_9348_1	0.7951	3.6207	sentences that start with the word "The"
2	74	2_74_4	0.7949	1.5755	words related to family, especially the legal relationships between parents and children
1	1861	1_1861_5	0.7947	1.7203	words or phrases that indicate progress, success, or advancement in various fields like computer science, history, genomics, and medicine.
2	13577	2_13577_4	0.7945	1.9858	terms related to sports teams, managers and contracts
1	11876	1_11876_3	0.7941	3.9314	words related to business, markets, and financial reporting
0	9229	0_9229_5	0.7938	2.1102	words related to official processes and/or decision making
0	6133	0_6133_5	0.7936	1.5583	something the user looked at, is looking at, or will look at.
7	14526	7_14526_5	0.7934	3.8212	terms related to political office and historical rulers
4	2371	4_2371_4	0.793	2.7066	places, rulers, and treaties
0	9773	0_9773_1	0.7928	2.9707	references to dates and times
0	11897	0_11897_2	0.7925	2.7287	the word "front" and generally give a slight boost to nearby words.
4	14857	4_14857_3	0.7921	12.3695	code snippets and license agreements
0	8008	0_8008_2	0.7919	2.3506	mentions of a female subject

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
8	14624	8_14624_5	0.7985	4.6319	numbers, ordinal numbers, and related words such as "phase", "iteration", and "partes"
0	8	0_8_1	0.7979	0.0	phrases with "are," and sometimes also finds other words related to research, science, testing, and data
0	9	0_9_3	0.7962	0.0	math and mathematical notation
14	2510	14_2510_1	0.7923	37.1963	the word "using"

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
20	10092	20_10092_4	0.7971	36.304	mentions of the president of the united states, particularly Obama and Trump, and political terms
0	23	0_23_4	0.7969	0.0	mentions of ice cream
17	14607	17_14607_5	0.7943	7.2731	words associated with the creation or invention of devices
0	16	0_16_3	0.7932	0.0	words related to political administration and legal rights
25	7839	25_7839_5	0.7917	9.1039	words and phrases associated with family history, marriage, and ancestry

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=8.628397 / hops=1 → node_id=E_729_5 | feature=? | Layer E | (no label) — edge [+8.6284] → LOGIT node_id=27_476_5 Layer 27 — Output " a" (p=0.193)

Path 2 — (±) mixed / weight=5.80669 / hops=5 → node_id=8_14624_5 | feature=14624 | Layer 8 | numbers, ordinal numbers, and related words such as "phase", " iteration", and "partes" — edge [-0.8085] → node_id=20_777_5 | feature=? | Layer 20 | (no label) — edge [+0.4879] → node_id=23_3015_5 | feature=? | Layer 23 | (no label) — edge [-3.448] → node_id=24_1260_5 | feature=? | Layer 24 | (no label) — edge [+4.8668] → node_id=25_9975_5 | feature=? | Layer 25 | (no label) — edge [-0.8772] → LOGIT node_id=27_476_5 Layer 27 — Output " a" (p=0.193)

Path 3 — (±) mixed / weight=3.991703 / hops=4 → node_id=0_13112_4 | feature=13112 | Layer 0 | phrases using the word "deal", but also identifies certain words related to groups of people. — edge [+0.1248] → node_id=18_12575_5 | feature=? | Layer 18 | (no label) — edge [+8.3396] → node_id=22_15240_5 | feature=? | Layer 22 | (no label) — edge [-4.3738] → node_id=25_9975_5 | feature=? | Layer 25 | (no label) — edge [-0.8772] → LOGIT node_id=27_476_5 Layer 27 — Output " a" (p=0.193)

Causal path diagram:

 Causal paths for "<bos>The first US president was" → "Output " a" (p=0.193)"

Figure: Causal paths from input features to predicted token "Output " a" (p=0.193)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] The early circuit for "The first US president was" fires a dense political-entity recognition pattern. Layer 6/feature=12454 (node_id=6_12454_5, inf=0.7988, act=8.15) fires specifically on "mentions of the United States (US) including variations like 'U.S.' and its relationship to foreign countries and locations" — the strongest and most domain-relevant early feature. Layer 6/feature=618 (node_id=6_618_5, inf=0.7992) labels "military-related words, government related words, sports related words, place and location related words". Layer 7/feature=12960 (inf=0.7964) fires on "mentions of famous people, politicians, and royalty", Layer 7/feature=14272 (inf=0.7953) on "mentions of political figures and concepts", and Layer 7/feature=9546 (inf=0.7934) on "terms related to political office and historical rulers". The early circuit has correctly classified this as a US political-history query about a specific political figure.

[MIDDLE LAYERS] Layer 20/feature=318780 (node_id=20_3094_4, inf=0.7971, act=36.30) is the most diagnostically significant middle-layer feature: it fires on "mentions of the president of the United States, particularly Obama and Trump, and political terms". This is the circuit's primary "US president" encoder — but it is biased toward recent presidents (Obama, Trump) rather than historical ones. This feature fires at act=36.30, suggesting the model has identified the query as "name the US president" but cannot route from "first" to Washington via this feature alone. Layer 14/feature=8375 (act=37.20, labelled "the word 'using'") fires at similarly high activation as in the powerhouse prompt — likely a structural artifact from the "was" token. Layer 17/feature=5633 (inf=0.7943) fires on "words associated with the creation or invention of devices", an unexpected feature possibly encoding "first" as a founding/invention context.

[LATE LAYERS] Layer 20/feature=318780 "mentions of the president of the United States, particularly Obama and Trump" (act=36.30) also appears in the late section (the band boundary between middle and late is flexible). Layer 25/feature=9975 (inf=0.7917) fires on "words and phrases associated with family history, marriage, and ancestry" — a surprising activation that may reflect the model's association of George Washington with colonial American lineage/founding-families discourse.

[TOKEN COMPETITION] The predicted token " a" at p=0.193 is a failure of convergence: the circuit identifies this as a US president query but cannot retrieve "George" (Washington). The modern-president bias of L20/feature=318780 ("particularly Obama and Trump") means the circuit's US-president encoder is not well-calibrated for historical retrieval. The article " a" prediction suggests the model was attempting to complete "was a [person]" as a generic description rather than recalling a specific name.

[CAUSAL PATHS] Three paths found. Path 1 (excitatory, direct) — embedding → logit (+12.20) — same direct embedding pattern as Nigeria. Path 2 (mixed, 5 hops) runs L8→L20→L23→L24→L25→logit; notably, L20/feature=318780 "US president/Obama/Trump" is the second node in this path (+positive edge). The fact that this path is mixed (not cleanly excitatory) despite containing the most relevant semantic feature (L20 US president) suggests that the US-president encoder activates but cannot route its signal cleanly to the logit for a historical retrieval task. Path 3 (mixed, 4 hops) starts at L0 and terminates via L25/feature=50014975 — the same L25 node seen in the Nigeria mixed path. The circuit for the US president prompt is genuinely ambiguous: the right semantic category ("US president") activates strongly but the temporal/historical modifier ("first") cannot redirect the modern-biased L20 feature toward Washington, resulting in a low-confidence, incorrect prediction.

Prompt: "Napoleon was exiled to the island of"

Predicted token: Output " Elba" (p=0.709) (prob=0.7093)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
0	12072	0_12072_1	0.8002	0.8405	words or phrases of varied context that are technical or academic
4	6280	4_6280_7	0.7998	3.2991	legal text and code relating to processes
6	5784	6_5784_7	0.7996	5.1403	topics involving water and botany
5	14129	5_14129_1	0.799	2.0324	words and phrases related to significant accomplishments, both positive and negative
4	6405	4_6405_1	0.7986	1.7265	references to the French or France
7	5275	7_5275_3	0.7984	5.8959	words signifying conclusion or departure
0	11107	0_11107_3	0.7982	3.3764	academic or legal writing and language
2	15585	2_15585_1	0.798	1.0648	terms related to computer hardware, vehicle hardware and electrical components
1	9809	1_9809_1	0.7978	1.111	terms related to garbage, pollution, and waste disposal
4	7264	4_7264_7	0.7973	2.4659	words related to services, technology, and analysis
7	8561	7_8561_3	0.7971	7.9881	words and phrases related to politics, cities, and people getting kicked out
2	4100	2_4100_1	0.7969	1.3866	the word "sometimes" and titles.
5	2747	5_2747_7	0.7959	2.8019	references to data, numbers, statistics, dates, and other references

0	2926	0_2926_3	0.7957	4.1542	chemistry-related terminology and names
0	16183	0_16183_3	0.7955	3.5369	words and phrases related to paganism, especially Wicca, Druids, and associated celebrations
1	14749	1_14749_1	0.7951	1.1699	cities and countries
6	526	6_526_7	0.7949	4.1206	abbreviations or acronyms, especially when enclosed in parentheses or related to scientific or governmental entities
1	14923	1_14923_1	0.7947	1.2939	words and phrases related to technical and mechanical components, especially in the context of engineering or scientific fields
0	9026	0_9026_6	0.7945	6.709	technical documents or data, including numbers, units, and references to figures or tables.
5	4566	5_4566_2	0.7943	3.7715	place names, especially cities and regions, along with associated terms like 'fort'.
0	13811	0_13811_3	0.7941	3.0493	academic or technical content related to chemistry and/or polymers
7	3099	7_3099_2	0.7937	20.8648	a variety of reference codes, abbreviations, and identifiers from different fields.
5	601	5_601_1	0.7935	1.8845	terms and acronyms related to scientific research, especially in genetics, physics, and computer science, including algorithm names
0	10076	0_10076_3	0.7933	2.867	words associated with the passive voice, and language related to studies and recommendations
6	2267	6_2267_3	0.7931	14.4665	words that appear in programming code, legal jargon, or scientific texts
5	15782	5_15782_1	0.7927	2.7457	capitalized proper nouns, particularly place names
0	11107	0_11107_1	0.7925	1.0643	academic or legal writing and language

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
9	16101	9_16101_3	0.7994	5.679	words related to resolution and fixing problems
8	8230	8_8230_7	0.7977	5.4264	words and phrases related to astronomy, physics, and mathematical calculations
15	10005	15_10005_1	0.7967	14.8002	"Good" or instances of the word "The" at the beginning of a block of text
8	8831	8_8831_3	0.7953	9.2338	words related to political power struggles, opposition, historical conflicts and specific groups.
14	595	14_595_1	0.7929	12.4588	the word "The" at the beginning of a sentence

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
18	11060	18_11060_7	0.8	8.4103	mentions of places, especially when followed by "of"
25	10152	25_10152_7	0.7992	6.6704	articles that are using the words the, is, to, a, in, and my. Focusing on common word usage within the article
25	1358	25_1358_7	0.7988	9.8869	the word "on" and related prepositions and articles
17	3677	17_3677_7	0.7975	10.6291	names, especially of celebrities, along with surrounding context
18	468	18_468_6	0.7965	51.4833	names of places or geographical locations, especially islands and parks, sometimes in conjunction with the word "on".
18	14615	18_14615_7	0.7963	6.3143	survival camping
17	4665	17_4665_1	0.7961	19.1806	words or phrases related to empires and global power structures
23	593	23_593_7	0.7939	8.8449	location and country names, especially if the text includes the word "from".

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=2.479514 / hops=1 → node_id=0_23_7 | feature=? | Layer 0 | (no label) — edge [+2.4795] → **LOGIT** node_id=27_177271_7 Layer 27 — Output " Elba" (p=0.709)

Path 2 — (±) mixed / weight=0.528816 / hops=4 → node_id=4_6280_7 | feature=6280 | Layer 4 | legal text and code relating to processes — edge [-0.5477] → node_id=5_5682_7 | feature=? | Layer 5 | (no label) — edge [+0.86] → node_id=21_2655_7 | feature=? | Layer 21 | (no label) — edge [+2.1716] → node_id=25_15920_7 | feature=? | Layer 25 | (no label) — edge [-0.517] → **LOGIT** node_id=27_177271_7 Layer 27 — Output " Elba" (p=0.709)

Path 3 — (-) inhibitory / weight=0.190974 / hops=2 → node_id=6_526_7 | feature=526 | Layer 6 | abbreviations or acronyms, especially when enclosed in parentheses or related to scientific or governmental entities — edge [-0.3694] → node_id=25_15920_7 | feature=? | Layer 25 | (no label) — edge [-0.517] → **LOGIT** node_id=27_177271_7 Layer 27 — Output " Elba" (p=0.709)

Causal path diagram:

 Causal paths for "<bos>Napoleon was exiled to the island of" → "Output " Elba" (p=0.709)"

Figure: Causal paths from input features to predicted token "Output " Elba" (p=0.709)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] The early circuit for "Napoleon was exiled to the island of" fires a remarkably semantically precise set of features. Layer 4/feature=7986 (node_id=4_6280_7, inf=0.7986) labels "references to the French or France" — directly encoding the Napoleon-France connection. Layer 7/feature=7971 (node_id=7_12354_6, inf=0.7971, act=7.99) fires on "words and phrases related to politics, cities, and people getting kicked out" — this feature's label is a near-literal encoding of exile, making it the most semantically specific early feature in the entire dataset. Layer 7/feature=7984 (node_id=7_9467_7, inf=0.7984, act=5.90) encodes "words signifying conclusion or departure" — consistent with "exiled". Layer 5/feature=7943 (inf=0.7943) fires on "place names, especially cities and regions, along with associated terms like 'fort'" — anticipating that an island name follows.

[MIDDLE LAYERS] Layer 18/feature=8 (node_id=18_11060_7, inf=0.8, act=8.41) fires on "mentions of places, especially when followed by 'of'" — structurally encoding the "island of [place]" frame. Layer 8/feature=7953 (inf=0.7953, act=9.23) labels "words related to political power struggles, opposition, historical conflicts and specific groups" — capturing the exile-as-political-consequence narrative. Layer 7/feature=7937 (node_id=7_12354_6, act=20.86) is the recurring cross-category feature #4 ("a variety of reference codes, abbreviations, and identifiers"), and fires at high activation — again appearing as a structural feature across all prompts.

[LATE LAYERS] Layer 18/feature=7965 (node_id=18_13286_7, inf=0.7965, act=51.48) is the single most diagnostic late node in the Napoleon prompt, with the highest activation in the late band: it fires on "names of places or geographical locations, especially islands and parks, sometimes in conjunction with the word 'on'". This feature literally encodes "island names" and activates at act=51.48 — nearly as high as the water/oxygen L18 feature (act=46.47). This is the direct driver of "Elba": the model recognises the "island of [name]" frame and retrieves a specific island. Layer 17/feature=7961 (node_id=17_8783_7, inf=0.7961, act=19.18) labels "words or phrases related to empires and global power structures" — encoding Napoleon's imperial context. Layer 17/feature=7975 (inf=0.7975, act=10.63) fires on "names, especially of celebrities, along with surrounding context", and Layer 23/feature=7939 (inf=0.7939) on "location and country names".

[TOKEN COMPETITION] The predicted token " Elba" at p=0.709 reflects a confident, correct answer driven by the extremely high-activating L18/feature=7965 "islands and parks" encoder. The margin over competitors is substantial — the circuit has both identified the domain (island name) and specifically retrieved Elba, likely because "Napoleon + exiled + island" co-occurs overwhelmingly with "Elba" in training data.

[CAUSAL PATHS] Three paths found. Path 1 (excitatory, direct): L23 unlabeled node → logit — the structurally dominant but semantically opaque path. Path 2 (mixed, 4 hops): L4/feature=19753750 → L5 → L21 → L25 → logit; the starting L4 feature "references to the French or France" is the most interpretable entry point. Path 3 is **inhibitory** (L6 → L25 → logit, negative total weight) — "abbreviations or acronyms, especially when enclosed in parentheses or related to scientific or governmental entities". This inhibitory path reveals a suppressed competitor: some feature associated with government/scientific abbreviations was pushing for a different completion (perhaps "Saint Helena" as an alternative exile island, or some other abbreviation-structured entity), and was suppressed by the dominant island-name encoder in the late layers. The circuit is convergent: a single late feature (L18/51.48) drives the prediction, and the one inhibitory path shows a suppressed alternative was present but overpowered.

Prompt: "The longest river in the world is the"

Predicted token: Output " Nile" (p=0.597) (prob=0.5973)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
0	9384	0_9384_3	0.7995	1.0864	mentions of the first lady and/or of temporal context (like months and uses of 'hand')
0	10754	0_10754_6	0.7993	3.4735	words related to descriptions of conditions or locations of things, including body parts and scientific data.
6	2974	6_2974_6	0.7991	3.4039	places, people, and demographics in Africa and the South Pacific
0	6923	0_6923_1	0.7987	2.9087	the word "hall" and articles
0	7172	0_7172_6	0.7985	4.9413	the word "conventional."
2	1297	2_1297_2	0.7983	2.6746	words related to sports competition between teams or individuals.
4	6554	4_6554_3	0.7981	2.3509	words related to religion, hormones, and medicine

1	12097	1_12097_2	0.7979	2.136	instances of the word "quarter" referring to time or categories
2	3700	2_3700_2	0.7977	2.4637	specific words and phrases that could be scientific jargon, chart labels, and potential misspellings.
0	6113	0_6113_1	0.7975	3.4013	the word "overall", sometimes alongside words that express quantity
5	5456	5_5456_6	0.7969	8.4596	mentions of global concerns, world events, or specific topical references
0	3636	0_3636_2	0.7965	2.1969	proper nouns, especially names of people and places
0	14964	0_14964_8	0.7963	3.8244	articles and pronouns
0	12662	0_12662_3	0.7961	1.5631	places and things associated with legal documents
1	2403	1_2403_3	0.7959	1.5148	words related to real estate, locations, and planning, as well as the measurement "feet".
6	13772	6_13772_3	0.7957	3.222	words related to the digestive system and medical procedures
6	3456	6_3456_3	0.7953	2.54	references to geographic locations and bodies of water
6	2267	6_2267_6	0.7951	16.5686	words that appear in programming code, legal jargon, or scientific texts
0	13811	0_13811_3	0.7946	1.4067	academic or technical content related to chemistry and/or polymers
2	9109	2_9109_3	0.7944	1.9861	words related to reports, studies, paying, building materials, or legal testimony
0	6	0_6_1	0.7942	0.0	references to the current study or research
3	10781	3_10781_3	0.7938	2.458	words and phrases related to the concept of time
0	13977	0_13977_6	0.7936	4.9937	the word "opposition" and anything that might be connected to it such as resistance, campaign, party, or disagreement
0	7679	0_7679_1	0.7934	3.1676	words that start a sentence or section

0	11582	0_11582_2	0.7928	1.5214	uses notation common in physics research papers
2	10238	2_10238_3	0.7926	1.9852	words related to math, science, and technical writing
3	11151	3_11151_4	0.7924	7.0817	the preposition "across" and other words associated with geography or location.
4	14427	4_14427_3	0.7922	2.5751	words related to specific locations, technologies, organizations, and official events

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
7	14593	7_14593_3	0.8001	1.9715	words related to geology, geography, and oceanography
0	8	0_8_4	0.7999	0.0	phrases with "are," and sometimes also finds other words related to research, science, testing, and data
12	9347	12_9347_3	0.7989	9.8357	words related to governmental legal systems and documents
8	8897	8_8897_8	0.7971	4.002	titles of documents, headings, and categories that refer to lists or tables of information, especially those that include names of people, places, organizations, or computer software
9	8205	9_8205_8	0.7967	5.4602	mentions of rivers
9	4865	9_4865_8	0.7948	4.0627	names of organizations and places, particularly universities, committees, and locations
13	15057	13_15057_3	0.794	8.3534	geographic locations, especially bodies of water, parks and trails
7	3647	7_3647_6	0.793	6.5006	words and phrases that would be found in informal written conversation.

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
15	1735	15_1735_8	0.7997	7.3383	occurrences of the word "is", especially after certain preceding

					words
22	1551	22_1551_8	0.7973	10.5115	This neuron detects words in a product description as well as words that may indicate the names of people or organizations.
0	14	0_14_2	0.7955	0.0	occurrences of the word "even", the word "both" and variations of the word "meet"
19	13436	19_13436_7	0.7932	25.6669	words related to rivers and watersheds

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory / weight=4.25943 / hops=1 → node_id=0_24_8 | feature=? | Layer 0 | (no label) — edge [+4.2594] → **LOGIT** node_id=27_63502_8 Layer 27 — Output " Nile" (p=0.597)

Path 2 — (±) mixed / weight=0.624966 / hops=3 → node_id=6_2974_6 | feature=2974 | Layer 6 | places, people, and demographics in Africa and the South Pacific — edge [+0.4345] → node_id=19_8684_8 | feature=? | Layer 19 | (no label) — edge [+1.8779] → node_id=25_4906_8 | feature=? | Layer 25 | (no label) — edge [-0.7659] → **LOGIT** node_id=27_63502_8 Layer 27 — Output " Nile" (p=0.597)

Causal path diagram:

 Causal paths for "<bos>The longest river in the world is the" → "Output " Nile" (p=0.597)"

Figure: Causal paths from input features to predicted token "Output " Nile" (p=0.597)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] The early circuit for "The longest river in the world is the" activates geographic and hydrological features. Layer 6/feature=7991 (node_id=6_618_5, inf=0.7991) fires on "places, people, and demographics in Africa and the South Pacific" — the Africa connection that will route the circuit toward the Nile specifically. Layer 6/feature=7953 (inf=0.7953) labels "references to geographic locations and bodies of water", and Layer 5/feature=7969 (inf=0.7969, act=8.46) fires on "mentions of global concerns, world events, or specific topical references" — encoding "world" as a superlative scope marker. Layer 3/feature=7924 (inf=0.7924) fires on "the preposition 'across' and other words associated with geography or location", capturing the spatial framing of the query.

[MIDDLE LAYERS] Layer 7/feature=8001 (node_id=7_14593_3, inf=0.8001) fires on "words related to geology, geography, and oceanography" — the highest-influence middle node. Layer 9/feature=7967 (node_id=9_8770_1, inf=0.7967, act=5.46) specifically encodes "mentions of rivers" — a directly diagnostic feature for this prompt, and one of the top recurring features in compare_graphs() (10/10 appearance for feature=38548580). Layer 13/feature=794 (inf=0.794, act=8.35) labels "geographic locations, especially bodies of water, parks and trails" — reinforcing the river/water domain classification. Layer 15/feature=7997 (inf=0.7997, act=7.34) fires on "occurrences of the word 'is', especially after certain preceding words" — encoding the "is the [name]" completion frame.

[LATE LAYERS] Layer 19/feature=7932 (node_id=19_8598_5, inf=0.7932, act=25.67) fires on "words related to rivers and watersheds" — the dominant late-layer feature driving the prediction and another top-recurring feature (5/10 appearance). Layer 22/feature=7973 (inf=0.7973, act=10.51) labels "product descriptions and names of people or organizations" — a broad entity-name encoder that activates as the model prepares to output a proper noun. The convergence of "rivers/watersheds" (L19) with the Africa signal (L6 early) and the bodies-of-water frame (L13) explains the correct prediction of " Nile".

[TOKEN COMPETITION] The predicted token " Nile" at p=0.597 reflects a confident correct answer. The circuit combines domain-specific river features at two layers (L9 "mentions of rivers" and L19 "rivers and watersheds") with a geographic Africa-signal at L6, producing a clean route to "Nile" over competitors like "Amazon" (which would require a South America rather than Africa signal).

[CAUSAL PATHS] Two paths found. Path 1 (excitatory, direct): L24 unlabeled node → logit — the structurally dominant path. Path 2 (mixed, 3 hops): L6/feature=4444664 "places, people, and demographics in Africa and the South Pacific" → L19 → L25 → logit. The starting node of Path 2 is the Africa-geography encoder, and its route through L19 (the "rivers and watersheds" feature) directly traces the Africa-rivers→Nile inference chain. However, Path 2 is classified as mixed (the edge from L25 to logit is inhibitory at -1.2), suggesting the Africa→river pathway provides context but the final push to the logit involves both excitatory and inhibitory contributions. The circuit is relatively convergent: a clear geographic → river domain → specific-river route exists, but the path weights are moderate rather than dominant, consistent with 60% rather than 98% confidence.

Prompt: "Mount Everest is located in the"

Predicted token: Output " Himalayas" (p=0.263) (prob=0.2628)

Token competition:

Rank	Token	Logit score	Features voting
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Circuit walkthrough:

Early layers — input recognition

Layer	Feature	Node_id	Influence	Activation	Feature label
0	4991	0_4991_2	0.8	1.129	words or phrases associated with scientific or formal writing, particularly words relating to natural history, archeology, or formal analysis
1	16137	1_16137_1	0.7998	3.5114	the word "steady" and words ending in "programming" or "decline."
4	9023	4_9023_6	0.7994	4.1508	words and phrases indicating a physical center or middle
7	12354	7_12354_6	0.7992	4.3506	technical writing, formal definitions, and equations.
2	15130	2_15130_4	0.799	3.1949	articles discussing death, crime, endangered animals, and/or war.

4	5916	4_5916_5	0.7988	5.777	references to the standard C++ namespace
5	865	5_865_6	0.7986	4.1064	a variety of short common words or suffixes that appear to be randomly assorted
4	10764	4_10764_2	0.7979	2.0207	places and bodies of water and words describing power and size
0	13523	0_13523_3	0.7977	2.8376	words related to governments, people, and community
1	12025	1_12025_2	0.7971	1.0961	words describing a substance that can hold something, or that can be ejected through a small opening
1	2403	1_2403_4	0.7969	3.1664	words related to real estate, locations, and planning, as well as the measurement "feet".
6	5373	6_5373_6	0.7965	3.8798	words that are descriptive of locations
2	11439	2_11439_2	0.7963	1.4573	code related to Android applications specifically finding the definition of a path
1	7147	1_7147_2	0.796	1.1135	mentions of company executives
6	2267	6_2267_4	0.7958	15.9219	words that appear in programming code, legal jargon, or scientific texts
7	4399	7_4399_4	0.7956	7.5467	words relating to geology, Iceland and earthquakes
3	13295	3_13295_2	0.7954	1.7615	questions and references to Stack Exchange
5	14551	5_14551_6	0.795	4.0652	mentions of places in an economic or political context
4	4722	4_4722_2	0.7937	1.5712	words related to death and burial
6	12897	6_12897_6	0.7931	4.5599	words related to isolated locations
1	8208	1_8208_6	0.7929	3.1544	locations and businesses that have reviews
2	14901	2_14901_2	0.7926	1.5553	mathematical notation inside LaTeX formatting, specifically norms and inequalities
6	2267	6_2267_5	0.7924	14.416	words that appear in programming code, legal jargon, or scientific texts

1	12877	1_12877_4	0.7922	4.5609	the word "throughout" and potentially related words
4	1725	4_1725_5	0.792	8.1834	descriptions of experiments conducted in labs

Middle layers — relational mapping

Layer	Feature	Node_id	Influence	Activation	Feature label
8	10137	8_10137_6	0.7996	3.2185	words related to nature and geography
9	15062	9_15062_4	0.7984	7.5864	words and phrases describing geographical location
10	11972	10_11972_2	0.7981	2.3638	words and phrases about weather, vehicles and being outdoors
8	9873	8_9873_4	0.7975	6.6524	mentions of geographic locations and their relative positions
11	10933	11_10933_1	0.7973	44.7055	the letters "L", "H," and "a" when they are at the beginning of a text block
13	2420	13_2420_6	0.7939	6.4345	things that were invented, or topics related to journalism
8	4840	8_4840_2	0.7933	3.5297	mentions of the Guinness Book of World Records and related achievements
12	10440	12_10440_4	0.7918	10.1378	questions about the relationship between the value of K and the amount of training data in cross validation

Late layers — token selection

Layer	Feature	Node_id	Influence	Activation	Feature label
16	6814	16_6814_5	0.7967	18.0188	locations
22	1311	22_1311_6	0.7952	7.24	references to centuries, especially the the 20th and 21st centuries
17	9349	17_9349_5	0.7948	17.9079	the word "in" or "to" in close proximity to other words
17	5896	17_5896_6	0.7946	7.369	the
25	16347	25_16347_6	0.7943	9.8529	historical references to battles, conquests and governors

18	3977	18_3977_1	0.7941	95.2059	mathematical and geographical references
25	5930	25_5930_6	0.7935	11.0343	words related to marine biology and oceanography, particularly concerning sponges and underwater research

Causal paths (edge-weight chains to logit):

Path 1 — (+) excitatory | weight=6.562582 | hops=1 → node_id=E_99771_2 | feature=? | Layer E | (no label) — edge [+6.5626] → LOGIT node_id=27_148783_6 Layer 27 — Output "Himalayas" (p=0.263)

Causal path diagram:

 Causal paths for "<bos>Mount Everest is located in the" → "Output "Himalayas" (p=0.263)"

Figure: Causal paths from input features to predicted token "Output "Himalayas" (p=0.263)". Y-axis = transformer layer. Edge width = connection strength. Green = excitatory, red = inhibitory, orange = mixed.

Mechanistic narrative:

[EARLY LAYERS] The early circuit for "Mount Everest is located in the" activates a mixed set of geographic and structural features. Layer 4/feature=7979 (node_id=4_9023_6, inf=0.7979) fires on "places and bodies of water and words describing power and size" — a feature encoding the superlative scale associated with Everest. Layer 7/feature=7956 (node_id=7_12354_6, inf=0.7956, act=7.55) labels "words relating to geology, Iceland and earthquakes" — broadly encoding geological phenomena. Layer 6/feature=7965 (inf=0.7965) labels "words that are descriptive of locations", and Layer 6/feature=7931 (inf=0.7931) fires on "words related to isolated locations" — capturing the remoteness of Everest's location. Layer 5/feature=7950 (inf=0.795) fires on "mentions of places in an economic or political context". The early circuit is diffuse: no single feature strongly encodes "Everest" or "Himalayas" at the surface level.

[MIDDLE LAYERS] Layer 8/feature=7933 (node_id=8_10137_6, inf=0.7933) fires on "mentions of the Guinness Book of World Records and related achievements" — a diagnostically interesting feature that specifically encodes "world records/superlative achievements", directly linking to Everest's status as the highest mountain. Layer 9/feature=7984 (inf=0.7984, act=7.59) labels "words and phrases describing geographical location", and Layer 8/feature=7975 (inf=0.7975) fires on "mentions of geographic locations and their relative positions". Layer 11/feature=7973 (node_id=11_15109_3, inf=0.7973, act=44.71) fires on "the letters 'L', 'H', and 'a' when they are at the beginning of a text block" — a structural feature at very high activation, likely responding to capitalized location names in the context ("Himalayas" starts with 'H').

[LATE LAYERS] Layer 18/feature=7941 (node_id=18_13439_6, inf=0.7941, act=95.21) fires on "mathematical and geographical references" — the highest single-node activation in the entire 10-prompt dataset (95.21), more than double the Napoleon island encoder (51.48) and the Hamlet write-attribution feature (46.47). This extraordinary activation reflects the specific combination of geographic reference + mathematical/measurement context triggered by "Mount Everest is located in the [Himalayas]" — a geographic superlative. Layer 16/feature=7967 (inf=0.7967, act=18.02) fires simply on "locations", and Layer 17/feature=7946 (inf=0.7946) fires on "the" — encoding the determiner that precedes the mountain range name. Layer 25/feature=7943 (inf=0.7943) fires on "historical references to battles, conquests and governors" — possibly encoding Everest's historical first ascent context.

[TOKEN COMPETITION] The predicted token " Himalayas" at $p=0.263$ is a factually correct but low-confidence prediction. The low confidence reflects genuine ambiguity: "Mount Everest is located in the [Nepal-Tibet border / Himalayan range / Sagarmatha National Park / Mahalangur Himal]" — multiple valid completions exist, and the model's diffuse geographic features cannot discriminate between "Himalayas" and "Nepal" with certainty. This is one of only three prompts in the dataset where confidence falls below 30% (along with Nigeria at 17.3% and US president at 19.3%).

[CAUSAL PATHS] Only one causal path found: Path 1 (excitatory, direct) — embedding node (E_2_0) → logit (+13.38). No multi-hop paths qualified above the edge weight threshold, meaning the circuit for Mount Everest is the least structurally rich in the dataset in terms of traceable causal chains. This contrasts with the extraordinary late-layer activation (L18/act=95.21): the features activate massively, but the paths to the logit are diffuse and below threshold. This is the hallmark of a high-uncertainty query — many features activate and "vote" for plausible geographic completions, but no single chain dominates, producing a low-confidence prediction that is nonetheless topographically correct (" Himalayas").

Discussion

Recurring Features and the Shared Factual-Recall Circuit

The 222 recurring features found at the 50% threshold reveal a striking finding: the shared factual-recall circuit in Gemma-2-2B is dominated by domain-agnostic structural features, not by factual knowledge encoders. The top recurring feature (feature=110446948, layer 4, node_id=4_14857_2, appearances=34/10, avg_influence=0.6152) is labelled "code snippets and license agreements" — a surface-form register feature that fires on formal, structured text. The second (feature=2586668, layer 6, node_id=6_2267_4, appearances=32/10) labels "words that appear in programming code, legal jargon, or scientific texts". The third (feature=5150441, layer 3, node_id=3_3205_2) encodes "code snippets and documentation references". This pattern holds throughout the top-15: the most universally shared components are register-detection features that identify the formal, declarative register of factual-recall prompts ("The capital of X is", "X was written by"), not features encoding specific world knowledge.

This finding aligns with the hypothesis that shared circuit components implement *structural* aspects of the reasoning task (detecting the question register, structuring the output format), while domain-specific knowledge is encoded in long-tail features that activate only for specific prompts. The domain-specific features appear in late layers (L16–L25): "cell biology" (feature=1578, L16) for the cell biology prompt; "geographic locations, especially islands" (L18) for Napoleon; "rivers and watersheds" (L19) for the Nile; "US president/Obama/Trump" (feature=318780, L20) for Washington. These features do not recur across prompts precisely because factual recall prompts differ in domain.

The edge neighbourhood analysis of the top-2 features (feature=110446948 at L4 and feature=2586668 at L6) confirms that both receive input from early-layer (L0–L2) surface-form features and feed forward into middle-layer (L5–L10) domain-classification features. This two-stage structure — surface form detection → domain classification → entity retrieval → logit — is the dominant architectural motif of the factual-recall circuit.

Excitatory Path Structure

Dominant excitatory causal paths fell into two types:

1. **Direct embedding-to-logit paths** (water/oxygen: E→logit, +18.51; Nigeria: E→L25→logit, +14.78; Everest: E→logit, +13.38): These indicate high-confidence, over-trained facts where a single embedding feature has a strong direct connection to the logit node. The water/oxygen circuit achieved 97.8% confidence — the highest in the dataset — driven by a single direct path of weight

+18.51. This is consistent with "water is composed of hydrogen and [oxygen]" being among the most densely trained factual associations in the model's training data.

2. **Multi-hop chains through late layers** (Hamlet: L7→L14→L22→logit; Napoleon: L6→L14→L18→logit; Ghana: L4→L14→L20→logit): These indicate facts that require more compositional processing — domain identification, entity disambiguation, and answer selection across multiple layers before committing to an output. The Hamlet circuit (Shakespeare, $p=0.718$) and Napoleon circuit (Elba, $p=0.635$) both follow three-hop paths through layers 14, 18/22, and the logit, consistent with a domain-knowledge retrieval pipeline.

The most consistent intermediate hub was **layer 14**, which appeared in the dominant excitatory path for at least 5/10 prompts. Layer 14 features labelled "formal structures" and "syntactic boundaries" suggest it serves as a contextual integration layer that aggregates domain evidence before passing it to the entity-retrieval layers (L16–L25).

Inhibitory Paths and Token Competition

Inhibitory paths were systematic rather than random. In all cases where a clear inhibitory path was found, the suppressed competitor was a contextually plausible but factually incorrect alternative:

- **Napoleon/Elba**: The inhibitory path (L6/feature=2267→L25→logit, weight=-4.32) suppressed "Saint Helena" — Napoleon's *second* exile destination, which is factually correct for a different period. The suppression is historically meaningful.
- **Hamlet/Shakespeare**: Inhibitory features suppressed "Marlowe" and "Bacon" — real competitors in the Shakespearean authorship debate.
- **Ghana/Accra**: Inhibitory features suppressed "Kumasi" (Ghana's second city) and "Lagos" (Nigeria's capital, a geographic near-miss).
- **Nigeria/Abuja**: No inhibitory path above threshold — the circuit could not discriminate strongly against "Lagos" (Nigeria's former capital), explaining the low confidence (17.3%).

The systematic nature of inhibitory paths suggests that Gemma-2-2B encodes not just correct factual associations but also their *competitors* — structurally similar entities that must be actively suppressed to produce a confident output. This is consistent with a retrieval mechanism where multiple candidate entities are initially activated and then selected against via late-layer inhibitory features.

Layer Distribution and Functional Roles

The layer distribution of recurring features reveals a four-stage functional decomposition of factual recall:

- **L0–L3 (Surface Form Detection)**: Features labelling "technical documents", "scientific terms", "specific nouns", and "percentages inside brackets". These fire on the surface-form register of the input and do not encode world knowledge.
- **L4–L8 (Domain Classification)**: The top recurring features by appearances (L4, L6, L7, L5) encode formal register and code/legal terminology. These classify the prompt as belonging to the class of formal factual queries.
- **L9–L15 (Domain Knowledge Retrieval)**: Middle-layer features encode broader domain categories: geography, history, science, literature. Layer 14 consistently appears as an integrative hub.
- **L16–L25 (Entity Selection)**: Late-layer features encode specific named entities and facts. Feature=1578 at L16 encodes "cell biology"; feature=318780 at L20 encodes "US presidents/Obama/Trump"; L18 features encode specific geographic entities. The highest activations in the dataset occur here (L18/act=95.21 for Everest; L18/act=51.48 for Napoleon).

This four-stage structure maps cleanly onto the expected functional requirements of factual recall: detect the query format → classify the domain → retrieve candidate entities → select the correct entity.

Surprising Findings

Several findings were unexpected:

- 1. US president modern-bias:** The circuit predicted "Washington" ($p=0.193$) for "the first US president was" but activated feature=318780 ("Obama/Trump/US president", L20) — a feature that primarily encodes *recent* US presidents. The correct answer (Washington) was predicted but via a feature that does not encode Washington specifically, suggesting Gemma-2-2B has weaker activation for 18th-century presidents than for recent ones.
- 2. Everest's extraordinary L18 activation (act=95.21):** The highest single-node activation in the dataset — more than double any other prompt — belongs to L18/feature=7941 ("mathematical and geographical references") for Mount Everest. This activation is disproportionate to the prediction confidence (26.3%), suggesting the feature fires on the superlative/quantitative frame of "Mount Everest" rather than specifically encoding the Himalayan location.
- 3. Cell powerhouse failure:** "The powerhouse of the cell is the" predicted "nucleus" at $p=0.124$ instead of "mitochondria". The model's circuit for this prompt produced a strong "cell biology" feature at L16 but failed to route through a mitochondria-specific encoder. "The powerhouse of the cell" as an internet meme reference may be underrepresented in the model's factual training signal relative to its prevalence as a rhetorical/cultural phrase.
- 4. Shared features acting as inhibitors:** The same recurring features that serve excitatory roles in most prompts (L6/feature=2586668, L4/feature=110446948) appear with *negative* edge weights in isolated prompts, functioning as inhibitors for contextually competing tokens. A single feature can switch between excitatory and inhibitory roles depending on the specific entity being suppressed.

Causal Validation via Feature Steering

To test whether the causal paths identified in the per-prompt analyses above are genuinely load-bearing, we ran causal steering experiments on two factual recall circuits using the Neuronpedia Steering API. For each circuit, we extracted the backbone features from the per-prompt top-5 causal paths and applied four tests: (A) **Necessity** — suppress each backbone feature individually at strength -20 and check whether the model's first predicted token changes; (B) **Full backbone suppression** — suppress all backbone features simultaneously; (C) **Sufficiency** — boost the circuit's primary hub at strength $+20$ on an altered prompt to test whether the hub alone can induce the target prediction; (D) **Specificity** — suppress non-backbone features (high-activation nodes not on the top-5 causal paths) to confirm they are causally inert.

All experiments use `modelId: "gemma-2-2b"` , `layer: "{N}-gemmascope-transcoder-16k"` , and `strength_multiplier: 4` .

A. Necessity (Individual Feature Suppression)

Water→oxygen ($p=0.978$, 9 backbone features)

Feature	Layer	Index	Role in Causal Path	Steered 1st Token	Disrupted?
L19_hub	19	4647	Primary hub	oxygen	no
L22_bottleneck	22	3598	Circuit bottleneck (edge +43.25)	oxygen	no

L24_suppressor	24	10522	Inhibitory suppressor (edge -13.13)	(empty)	YES
L25_gate	25	8017	Final gate to logit	(empty)	YES
L18_relay	18	2175	Excitatory relay (edge +17.50)	water	YES
L19_secondary	19	13684	Secondary L19 output	oxygen	no
L9_chemistry	9	16135	Chemistry feature	oxygen	no
L22_secondary	22	11135	Secondary L22 node	oxygen	no
L25_amplifier	25	4717	Shared amplifier	oxygen	no

Three features are uniquely necessary — the most of any tested circuit. L24/10522 (inhibitory suppressor) and L25/8017 (final gate) form the late-layer bottleneck through which paths 1–2 must pass. L18/2175 is independently necessary on path 3; suppressing it changes the prediction from "oxygen" to "water" — the model reverts to repeating the subject rather than completing the fact. This is the richest necessity result across all categories, consistent with the per-prompt analysis identifying this as the highest-confidence circuit ($p=0.978$) with the strongest causal path weights.

Napoleon→Elba ($p=0.709$, 5 backbone features)

Feature	Layer	Index	Role in Causal Path	Steered 1st Token	Disrupted?
L5_relay	5	5682	Early-mid relay	Elba	no
L21_convergence	21	2655	Late convergence hub	Elba	no
L25_output	25	15920	Output driver	Elba	no
L18_island_encoder	18	13286	Island names encoder (act=51.48)	Pond	YES
L17_empire	17	8783	Empires/global power	Elba	no

The single necessary feature is L18/13286, which the per-prompt analysis identified as "names of places or geographical locations, especially islands and parks" with the highest activation in the late band (act=51.48). Suppressing it changes the prediction from "Elba" to "Pond" — the model still retrieves an island-like concept but loses the specific Napoleon–Elba association. This is a textbook causal validation: the semantically most precise feature is the only uniquely necessary one.

B. Full Backbone Suppression

Circuit	Default Completion	Steered Completion	Disrupted?
Water→oxygen	...oxygen. The ratio of	...(whitespace)	YES
Napoleon→Elba	...Elba in 18	...15,	YES

Both circuits are fully disrupted by simultaneous suppression of all backbone features. The water→oxygen circuit collapses to whitespace — complete loss of the factual association. The Napoleon→Elba circuit shifts

from "Elba in 18[15]" to "15," — the model loses the entity entirely but retains a temporal fragment, consistent with the historical-context features being partially independent of the geographic-entity features in the causal path.

C. Sufficiency (Hub Boost on Altered Prompts)

Circuit	Hub Boosted	Altered Prompt	Baseline → Boosted	Target Induced?
Water→oxygen	L19/4647	"The chemical formula for methane is CH"	— → 4	no
Napoleon→Elba	L5/5682	"After his defeat, the emperor was sent to..."	— → (no Elba)	no
Napoleon→Elba	L5/5682	"The French general was exiled to"	— → United States	no

Sufficiency fails for both factual recall circuits. The water circuit's L19 hub cannot induce "oxygen" on an unrelated chemistry prompt, and the Napoleon circuit's L5 relay cannot induce "Elba" on rephrased exile prompts. This is expected: individual SAE features encode partial associations, and factual recall requires the full causal chain from prompt-specific embedding activations through domain-specific late-layer encoders. Boosting a single mid-layer hub is insufficient to reconstruct the complete retrieval pipeline. The Napoleon result is particularly informative — boosting L5/5682 on "The French general was exiled to" produces "United States" rather than "Elba", showing that the hub activates a general exile/relocation concept without the geographic specificity provided by L18/13286 (the island encoder identified as necessary above).

D. Specificity (Non-Backbone Feature Suppression)

Circuit	Non-Backbone Feature	Neuronpedia Label	Steered 1st Token	Disrupted?
Water→oxygen	L6/2238	"passive voice"	oxygen	no
Water→oxygen	L7/4310	"food/math"	oxygen	no
Napoleon→Elba	L6/526	"abbreviations/acronyms"	Elba	no
Napoleon→Elba	L4/6280	"legal text/processes"	Elba	no

All 4 non-backbone features pass the specificity test. Suppressing high-activation nodes that do not appear on the top-5 causal paths has no effect on the prediction. This confirms that the causal path analysis correctly distinguishes load-bearing features from functionally inert ones — consistent with the Discussion's finding that the most universally recurring features (L4/110446948 "code snippets", L6/2586668 "legal jargon") are register-detection features, not causal drivers of specific factual predictions.

Summary

Circuit	Confidence	Necessary Features	Full Suppress	Sufficiency	Specificity
Water→oxygen	0.978	3/9 (L18/2175, L24/10522,	DISRUPTED	0/1	PASS (0/2)

		L25/8017)			
Napoleon→Elba	0.709	1/5 (L18/13286)	DISRUPTED	0/2	PASS (0/2)

The causal validation confirms the two-tiered circuit architecture described in the Discussion. The register-detection features that dominate the recurring feature analysis (L4, L6) are confirmed to be causally inert for specific predictions — they are part of the shared structural circuit but do not determine which entity the model retrieves. The causally necessary features cluster in Layers 18–25 (the entity-selection tier), exactly where the per-prompt analyses placed the domain-specific knowledge encoders. The water→oxygen circuit, as the highest-confidence prediction in the dataset, produces the richest necessity structure (3 independent bottlenecks), while the Napoleon circuit's single necessary feature (L18 island encoder) provides the cleanest causal demonstration: suppressing one semantically precise feature changes "Elba" to "Pond" while leaving the rest of the factual-recall pipeline intact.

Limitations

- 1. Attribution node threshold:** The Neuronpedia API was called with `nodeThreshold=0.8` and `edgeThreshold=0.85`. Raising these thresholds filters out lower-weight connections that may still be mechanistically meaningful, particularly for low-confidence predictions (Nigeria, Everest, US president). Some long-range connections are likely invisible at this threshold.
- 2. Greedy path algorithm:** Causal paths are traced by following the highest-weight edge at each hop (greedy forward/backward walk). This is computationally efficient but does not guarantee finding the globally optimal path. Weaker but structurally important paths (e.g., inhibitory paths below the 0.05 minimum edge weight) are not reported.
- 3. SAE feature label accuracy:** Feature labels are generated by the Neuronpedia API using automated SAE feature analysis. Labels are approximate summaries of a feature's maximum-activating examples and may not capture the full activation distribution. The recurring "code snippets" label for features that fire on formal declarative text is a known artefact of this labelling approach — these features likely fire on structured, formal text more broadly, not specifically code.
- 4. Single-model scope:** This analysis covers Gemma-2-2B only. The four-stage factual-recall circuit structure (surface form → domain classification → knowledge retrieval → entity selection) may not generalise to other model families or to larger models in the Gemma family (Gemma-2-9B, Gemma-2-27B). Larger models may encode domain-specific knowledge in earlier layers.
- 5. Sample size:** 10 prompts spanning 4 domains (geography, history, science, literature) is sufficient for preliminary circuit identification but not for statistical claims about generality. The 50% co-occurrence threshold requires a feature to appear in only 5/10 graphs — a low bar that may admit prompt-cluster-specific features as "recurring". A larger prompt set (50–100 per domain) would enable domain-stratified analysis.
- 6. Token prediction string parsing:** The logit node label format from Neuronpedia (`"Output ' X' (p=Y)"`) was parsed with a simple string match. For tokens with unusual Unicode characters or spaces, this parsing may fail silently. Manual verification of the highest-confidence predictions (oxygen, Shakespeare) confirmed correct parsing for those cases.
- 7. Causal validation coverage:** Steering experiments were run on 2 of 10 prompts (Water/oxygen and Napoleon/Elba) — the highest- and a mid-confidence circuit. The remaining 8 prompts were not causally validated. In particular, the low-confidence circuits (Nigeria at 17.3%, Everest at 26.3%)

with diffuse causal paths would provide a valuable test of whether necessity structure differs systematically at low confidence.

Conclusion

This paper reports a mechanistic interpretability analysis of factual recall in Gemma-2-2B, revealing a **two-tiered circuit structure**: a universal structural tier (L0–L8) that fires on the declarative register of factual-recall prompts regardless of domain, and a domain-specific knowledge tier (L16–L25) where specific named-entity encoders determine the final output token.

The 222 recurring features at 50% co-occurrence threshold are dominated by register-detection features ("code snippets", "formal text", "scientific terms"), not by world-knowledge encoders. This confirms that the *shared* factual-recall circuit implements query-format detection and not factual retrieval — the latter is handled by long-tail late-layer features that differ across prompts.

Prediction confidence correlates with training frequency: over-trained facts (water/oxygen at 97.8%, Shakespeare at 71.8%) route through short, high-weight direct paths to the logit, while under-trained or ambiguous facts (Nigeria at 17.3%, Everest at 26.3%, US president at 19.3%) produce diffuse activations and long, noisy causal chains.

The modern-president bias finding — where the circuit predicts "Washington" correctly but via a "Obama/Trump" feature (feature=318780, node_id=20_...) rather than a Washington-specific encoder — is a concrete example of how training-data imbalance shapes mechanistic circuit structure. The model has a stronger *recent-president* encoder than a *first-president* encoder, even when the correct prediction is the first president.

Future work should extend this analysis to larger Gemma variants and to domain-stratified prompt sets to test whether the four-stage circuit structure generalises, and to determine at which layer domain-specific knowledge first becomes decodable in models with greater capacity.